

Honors Thesis - NICU Data Analysis

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Data Dictionary

Variable	Description
patient_id	De-identified, randomized patient identification number assigned to each infant
survival_status	Whether the infant was alive or deceased at the time of discharge
gestational_age	Gestational age at birth (in weeks)
birth_weight	Birth weight of the infant (in grams)
gender	Sex of the infant
inborn	Whether the infant was born at Duke Hospital or transferred from another facility (Yes , No)
race_ethnicity	Racial/ethnic classification of the infant (White , Black , Other)
length_of_stay	Total length of stay (in days) from admission to discharge or death
ivh	Presence or grade of intraventricular hemorrhage (IVH), bleeding into the internal structures of the brain (None , Grade I , Grade II , Grade III , Grade IV)
rop	Presence or stage of retinopathy of prematurity (ROP), a condition affecting blood vessels in the eye that leads to blindness if not promptly treated (None , Stage I , Stage II , Stage III-IV (No Cryo or Laser Surgery) , Stage III-IV (With Cryo or Laser Surgery))
nec	Presence or type of necrotizing enterocolitis (NEC), a severe neonatal intestinal condition (None , Medical , Surgical)
pda	Presence and management of patent ductus arteriosus (PDA), a neonatal cardiovascular anomaly requiring medical or surgical intervention (None , None (Medication Only) , Yes (Medically Treated) , Yes (Ligated) , Yes (Untreated))

Data Cleaning

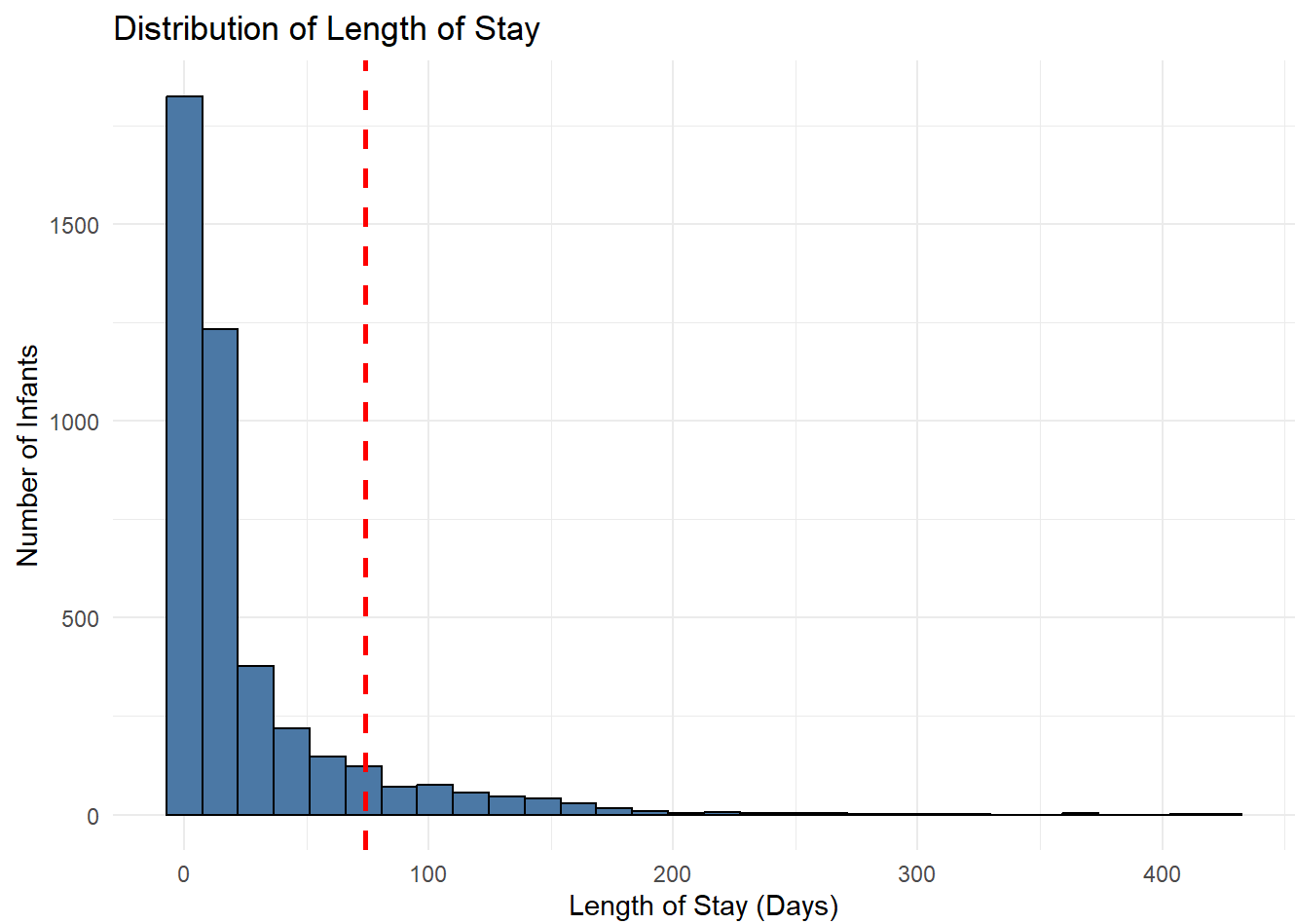
##	patient_id	survival_status	gestational_age	birth_weight	gender
##	0	0	0	0	0
##	inborn	race_ethnicity	length_of_stay	ivh	rop
##	0	0	53	0	0
##	nec	pda			
##	0	0			

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	22.00	31.00	34.00	34.01	38.00	43.00

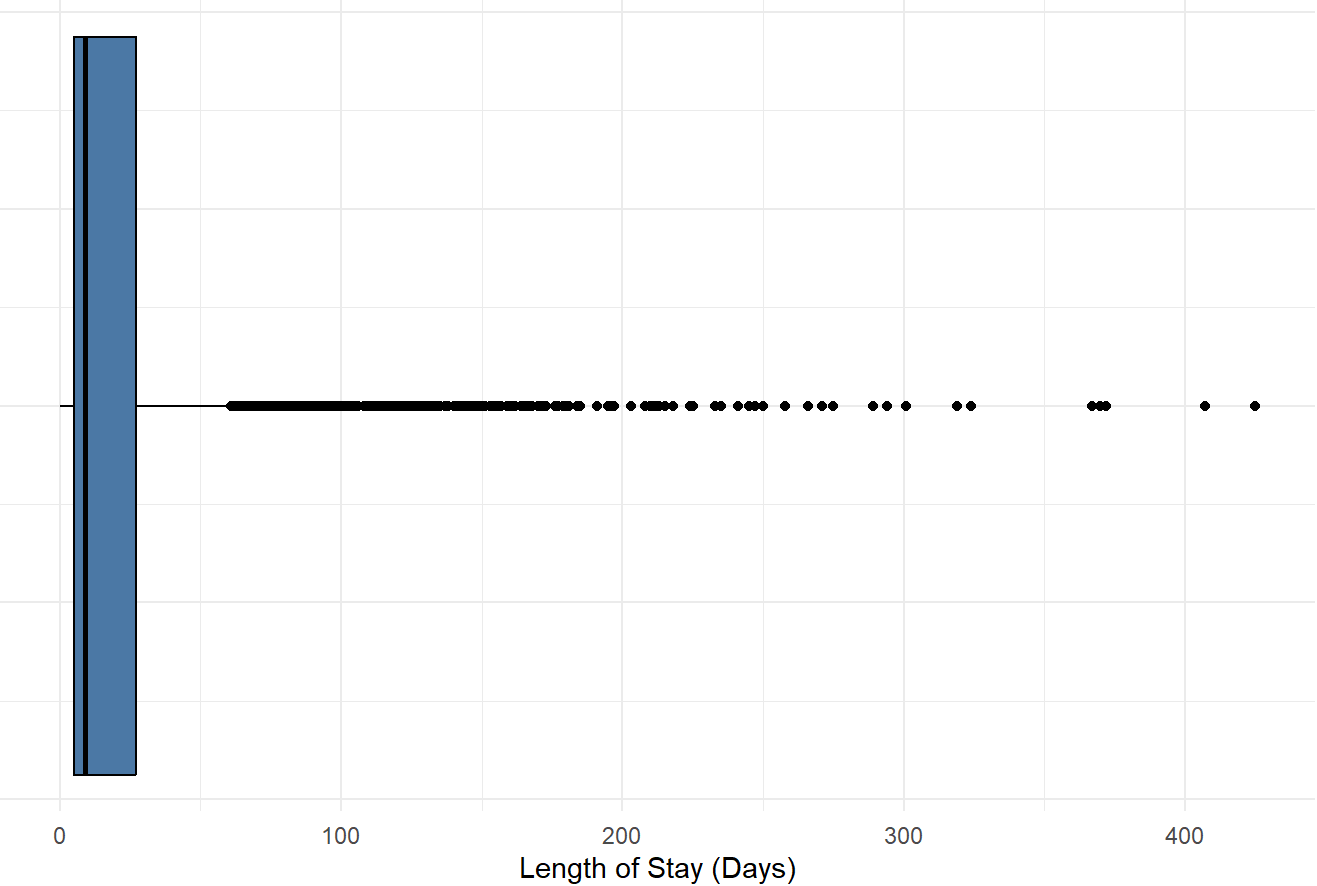
##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	280	1521	2255	2309	3100	5870

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
##	0.0	5.0	9.0	25.7	27.0	425.0	53

Exploratory Data Analysis



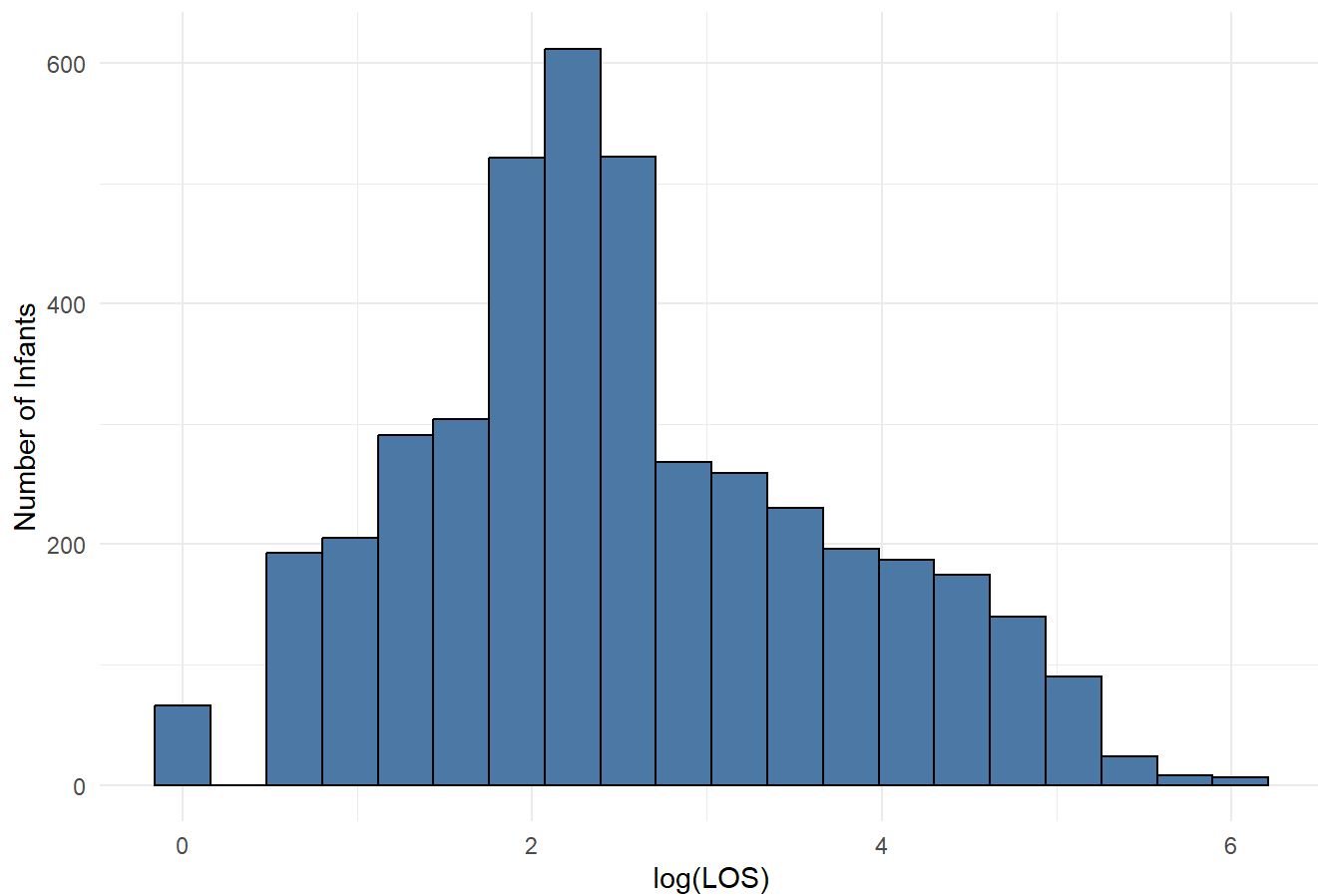
Distribution of Length of Stay



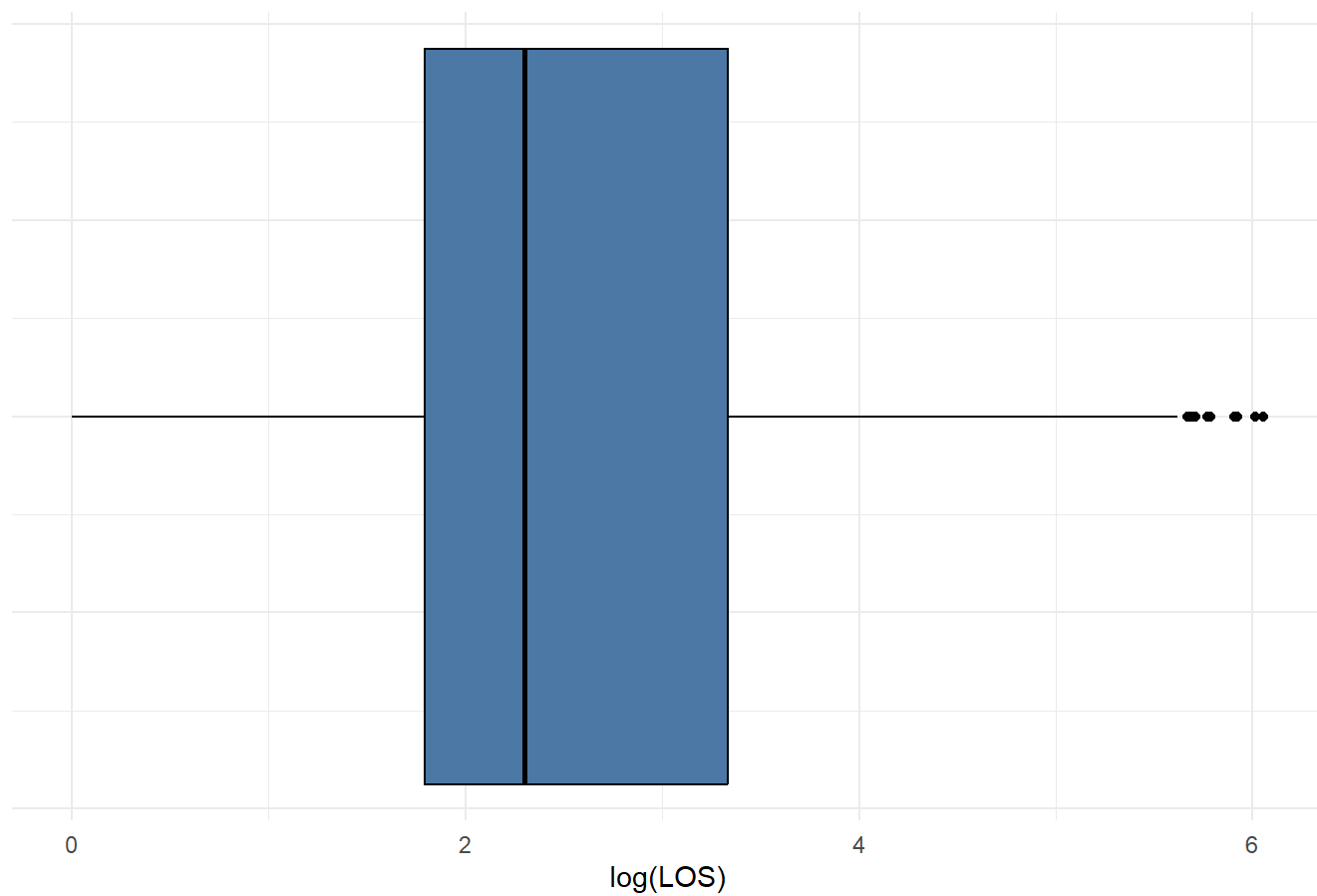
```
## # A tibble: 1 × 7
```

##	n	mean	sd	median	Q1	Q3	IQR
##	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	4297	25.7	40.8	9	5	27	22

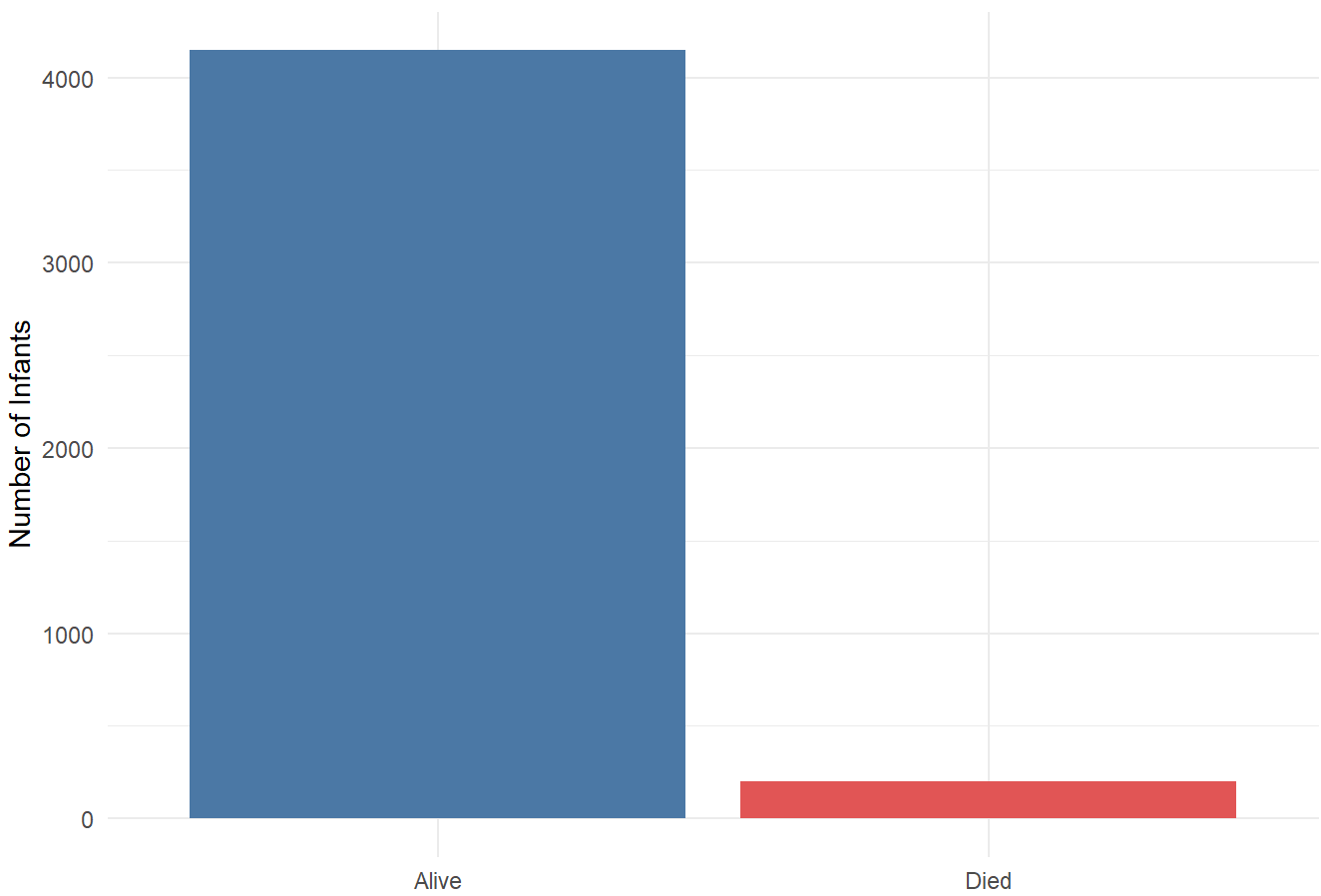
Log-Transformed Distribution of Length of Stay



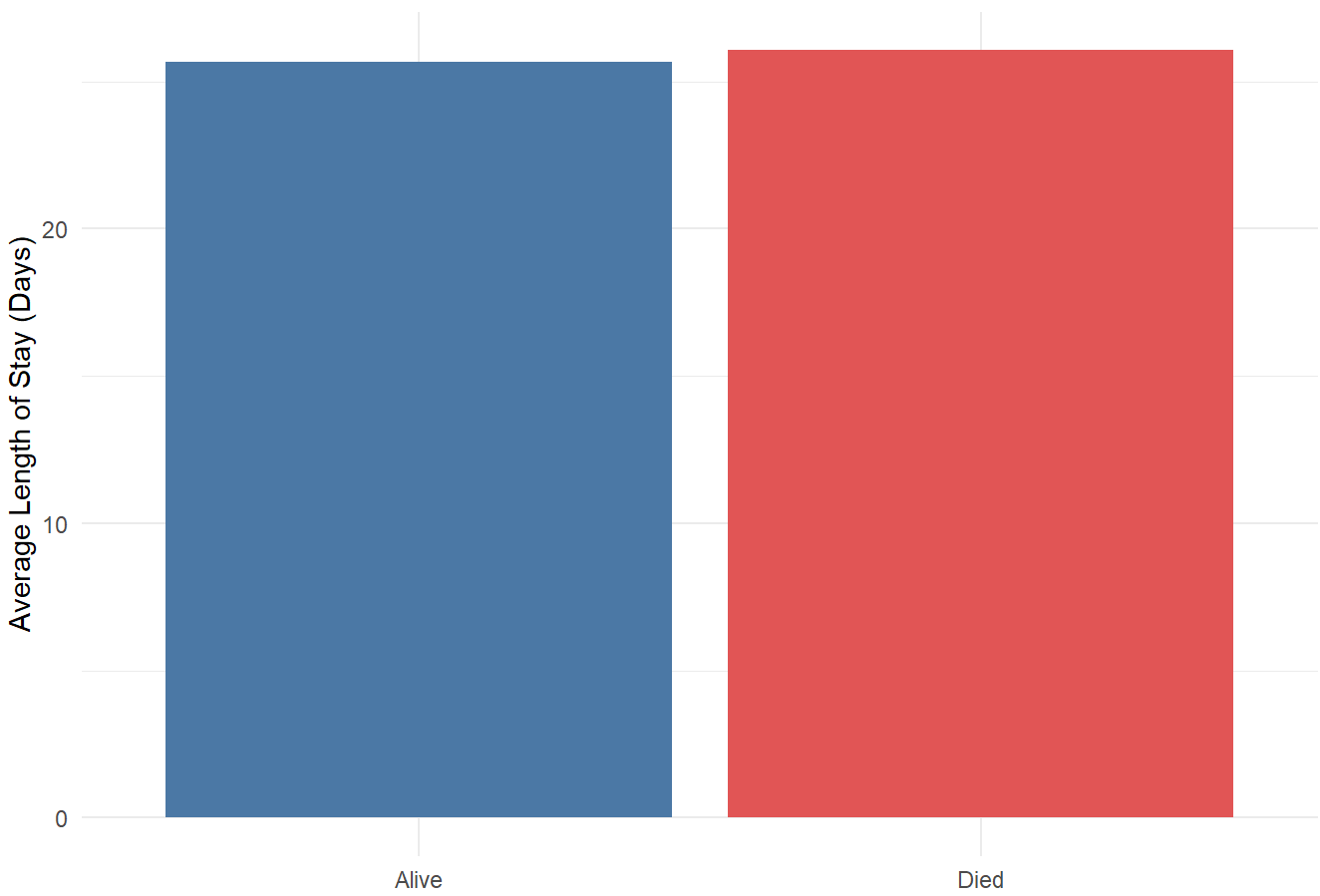
Log-Transformed Distribution of Length of Stay



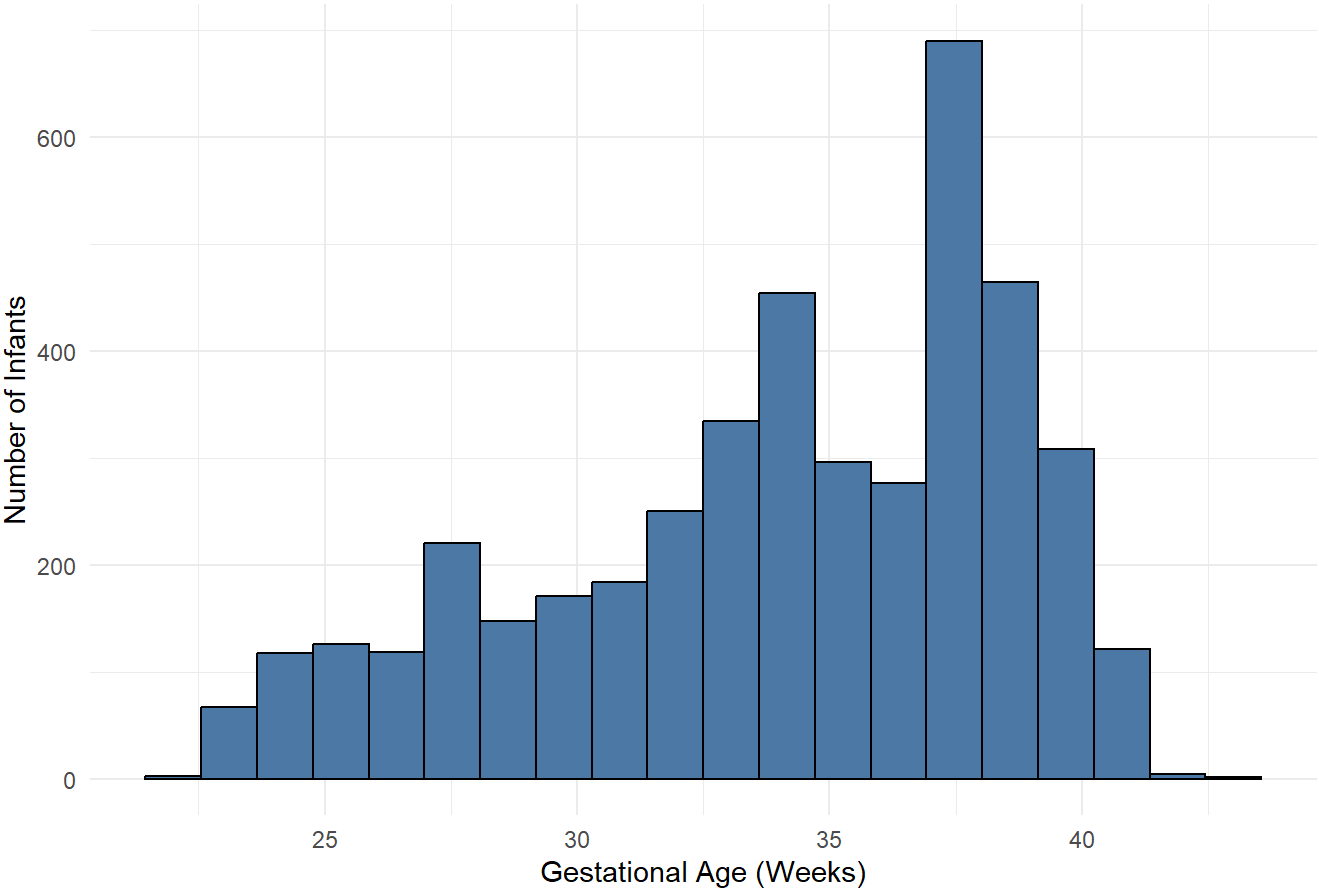
Distribution of Survival Status



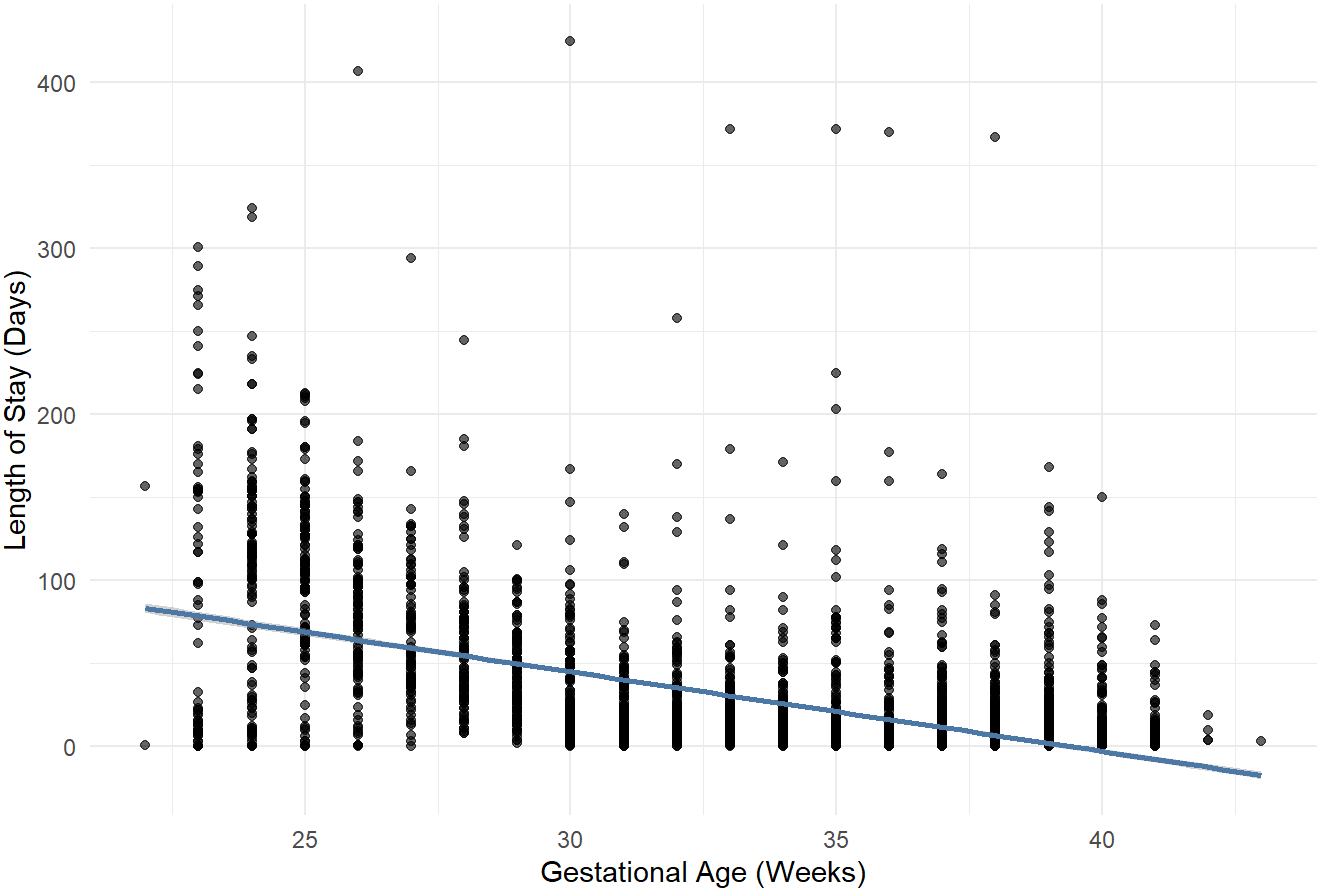
Average Length of Stay by Survival Status



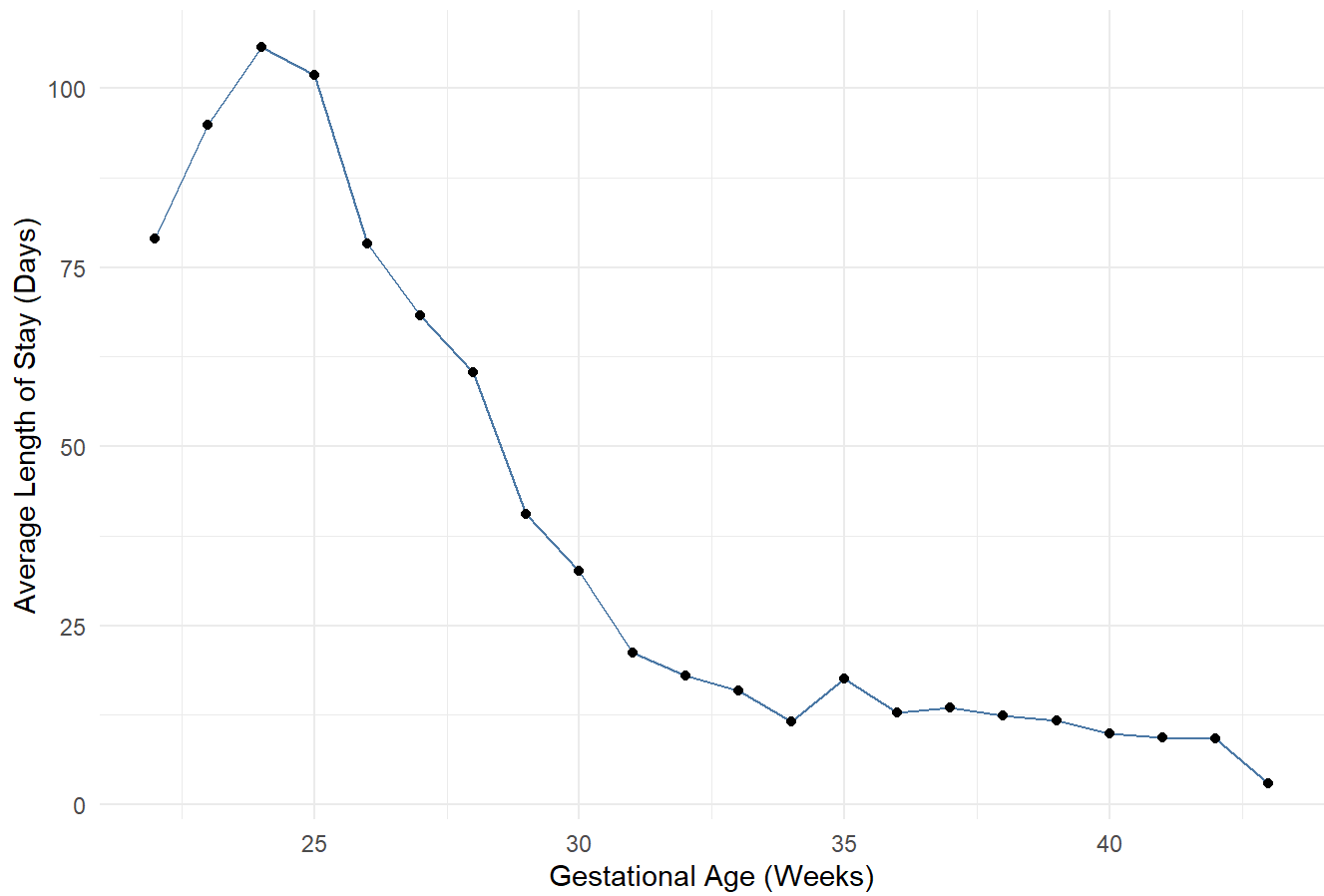
Distribution of Gestational Age



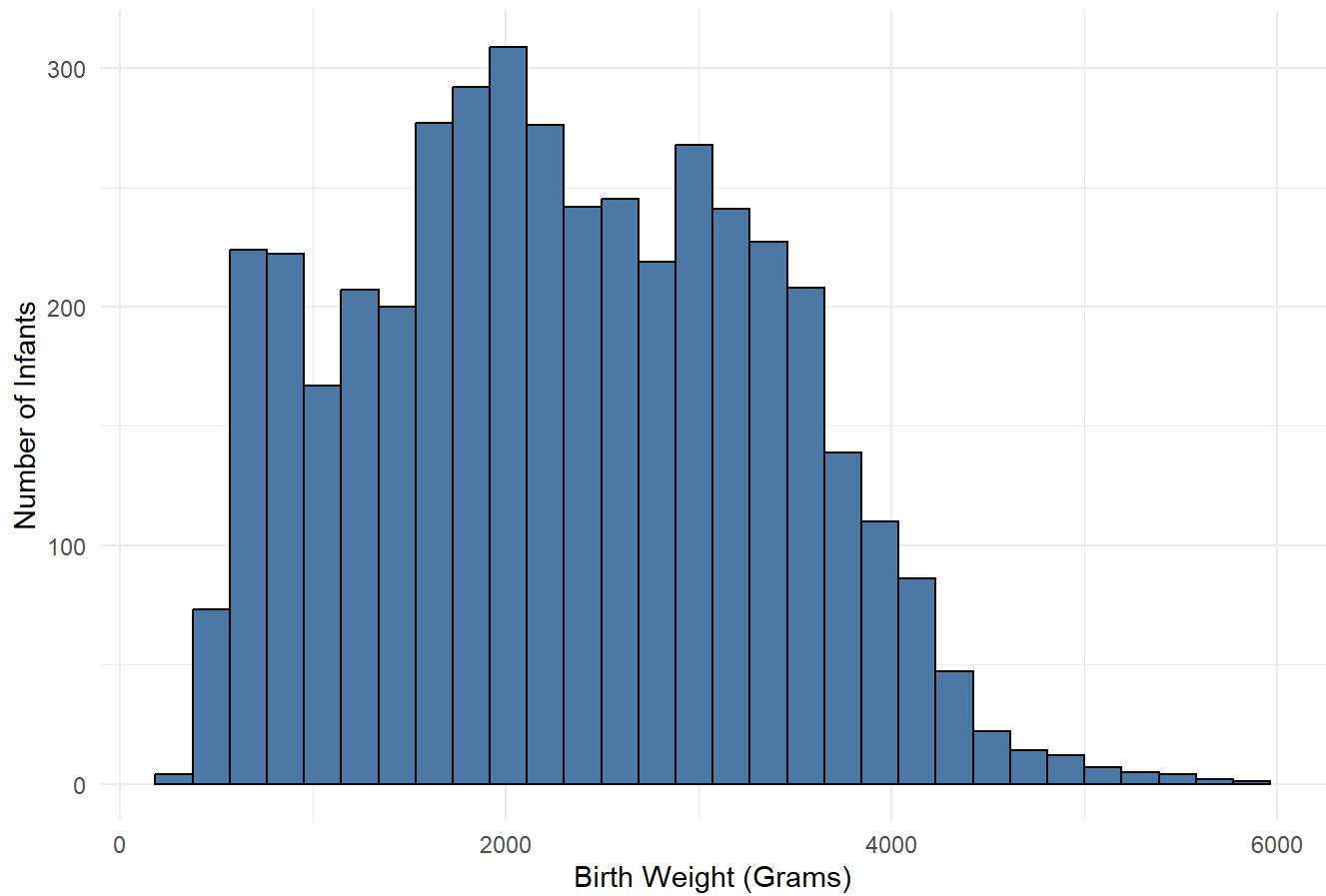
Length of Stay vs. Gestational Age



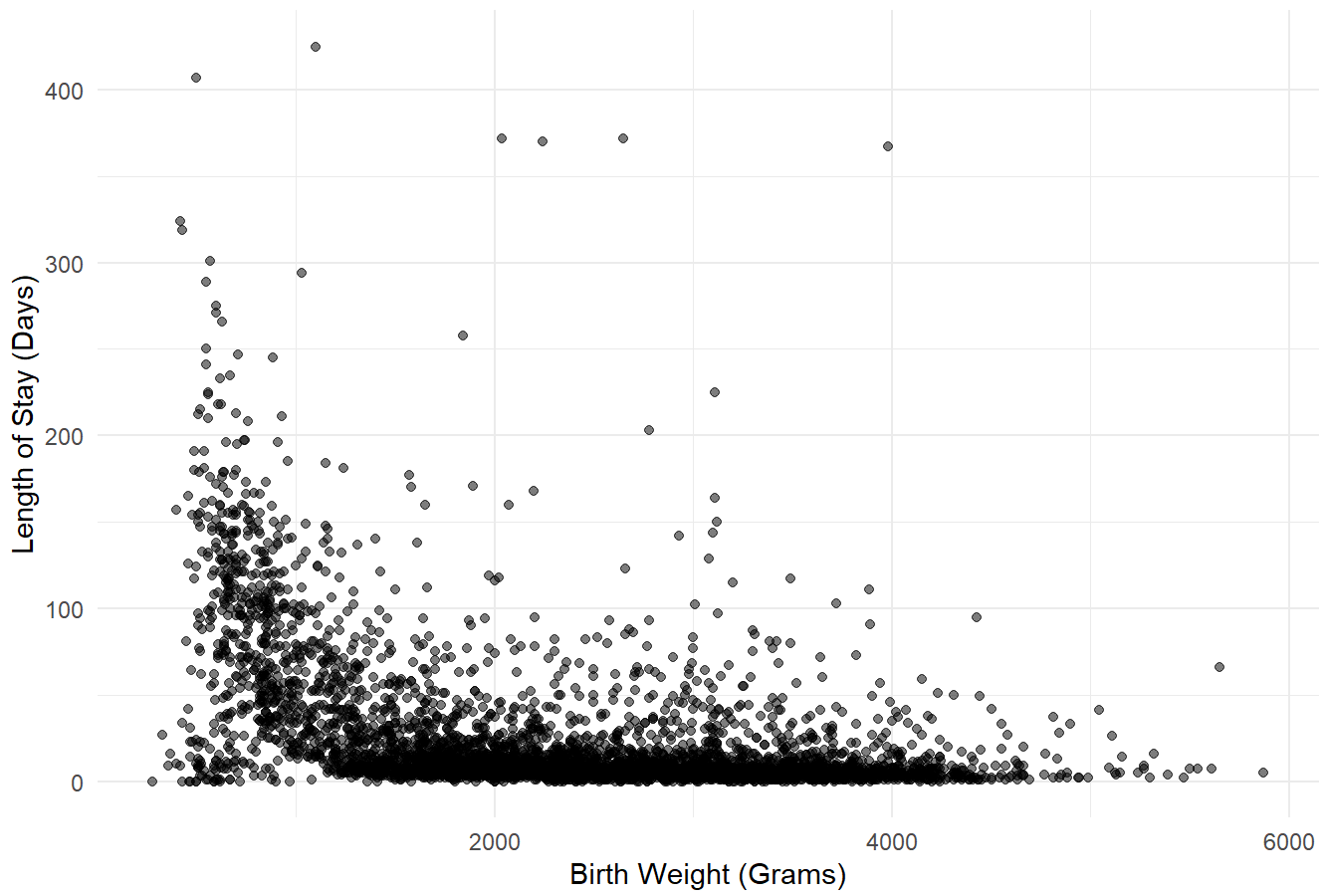
Average Length of Stay by Gestational Age



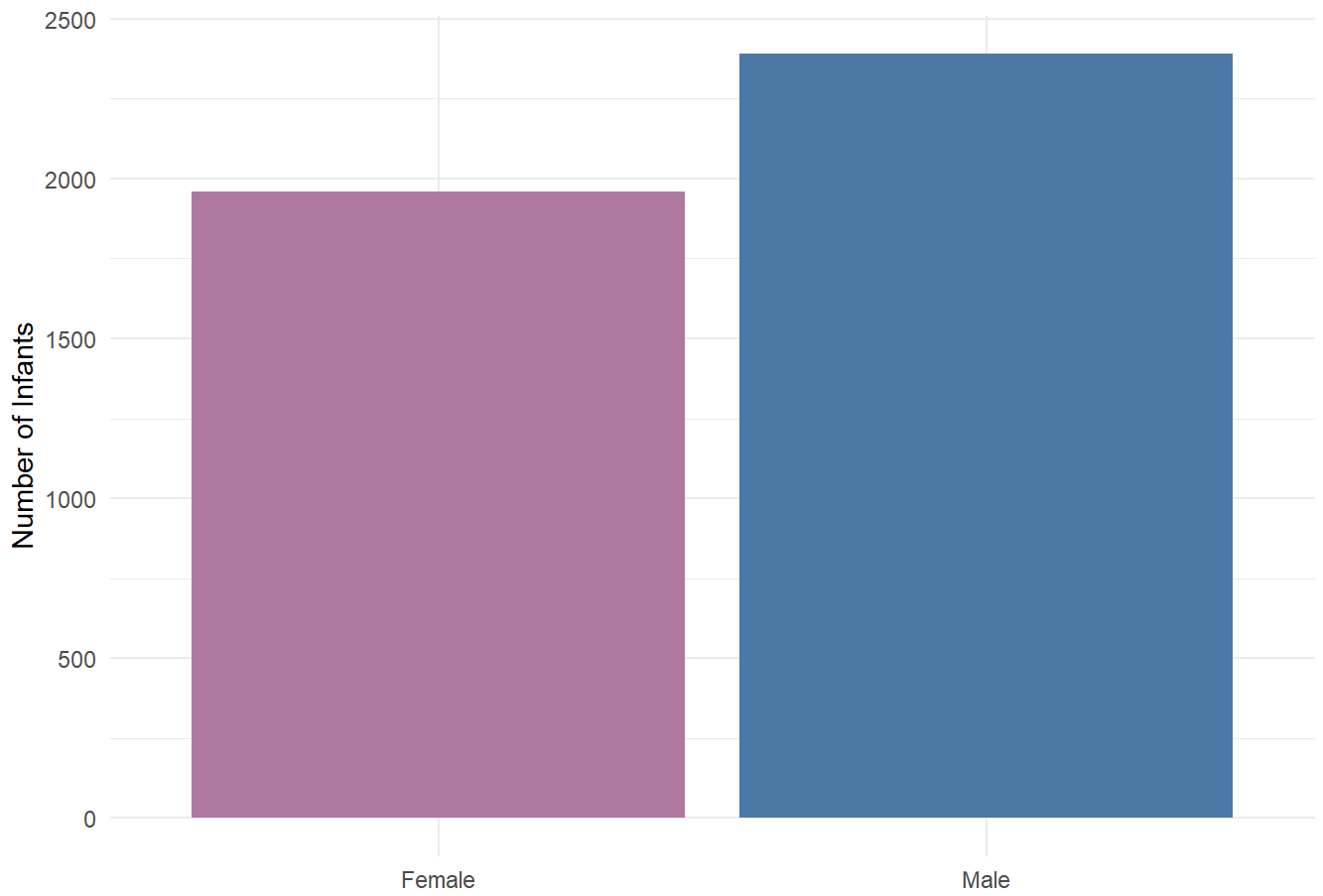
Distribution of Birth Weight



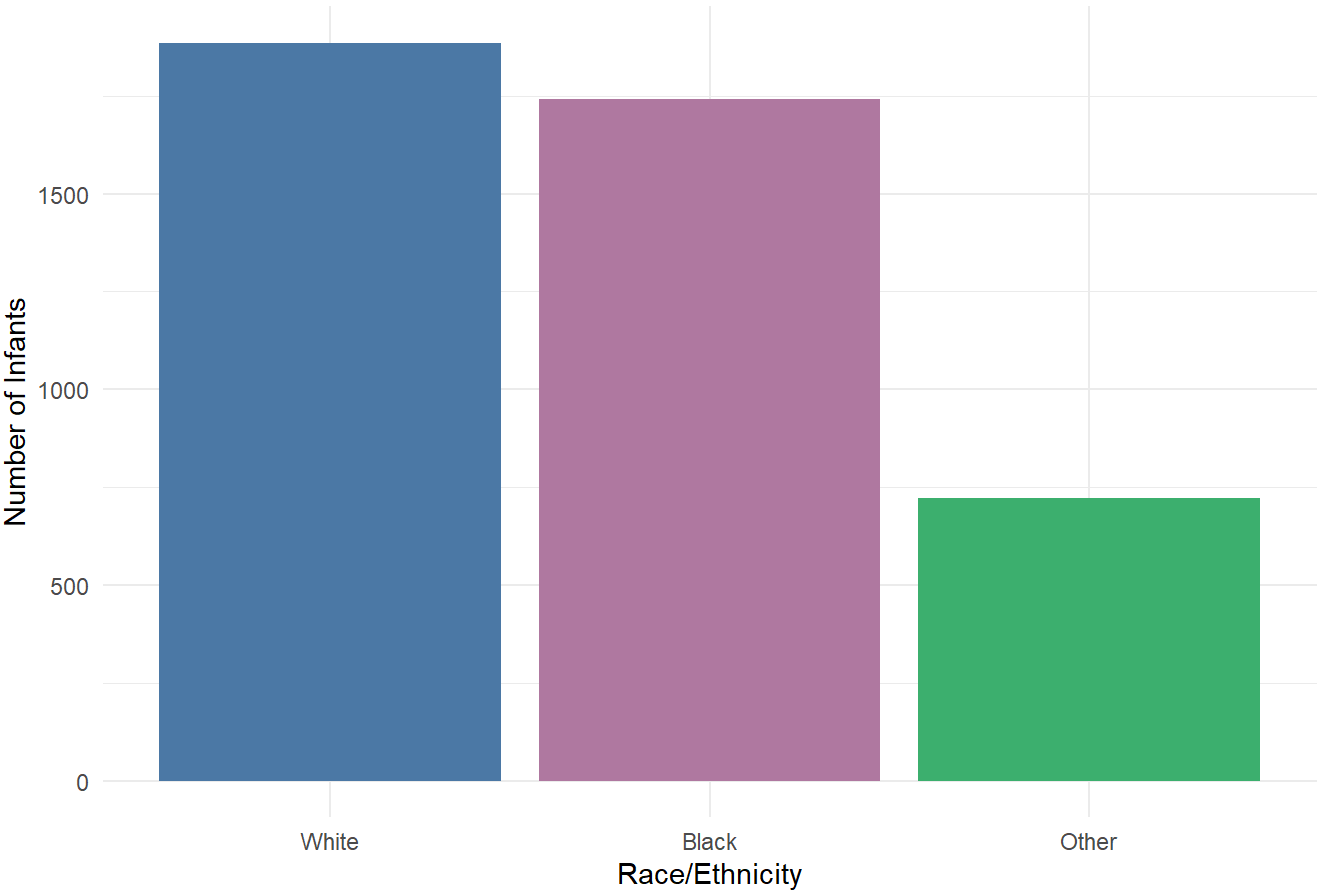
Length of Stay vs. Birth Weight



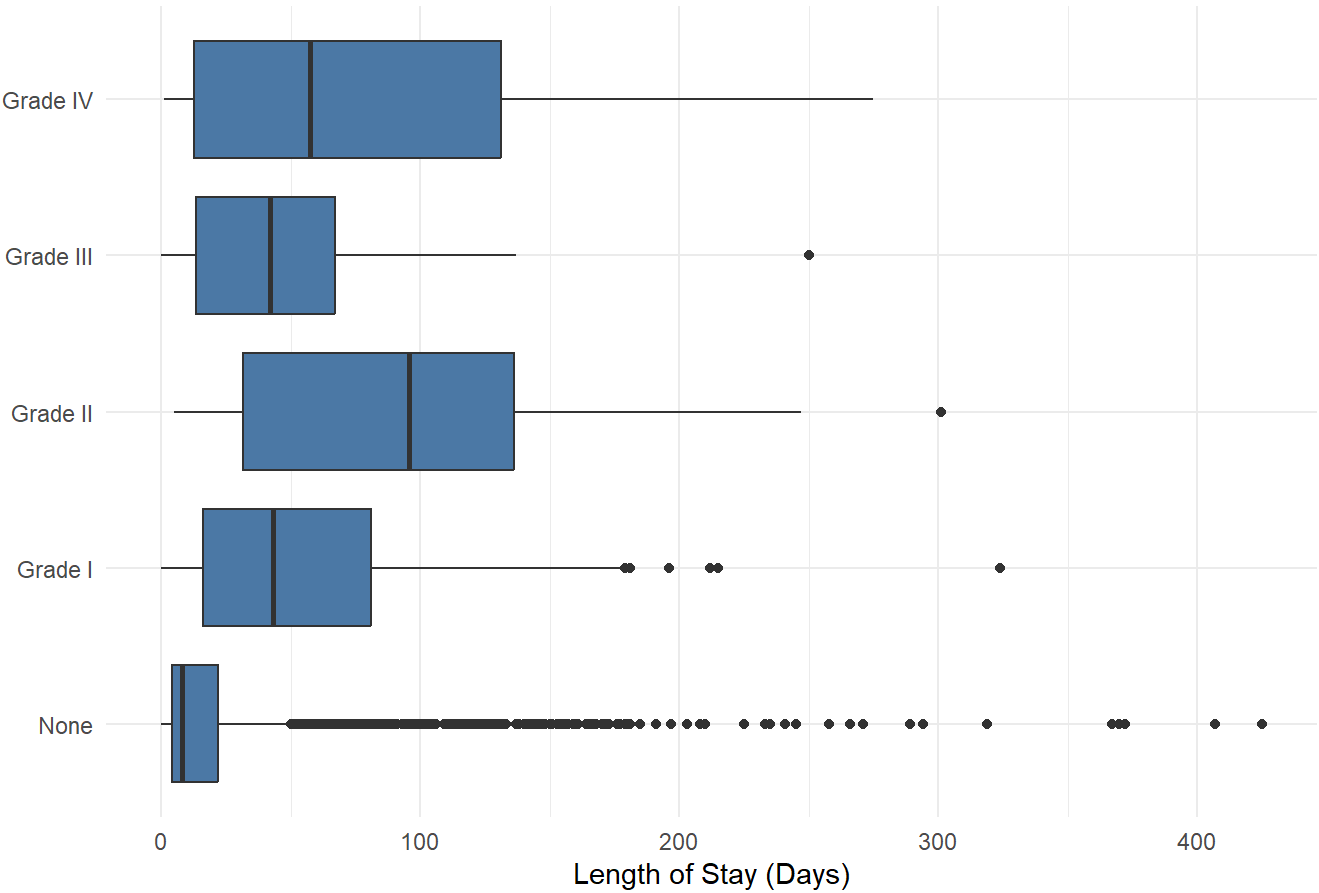
Distribution of Gender



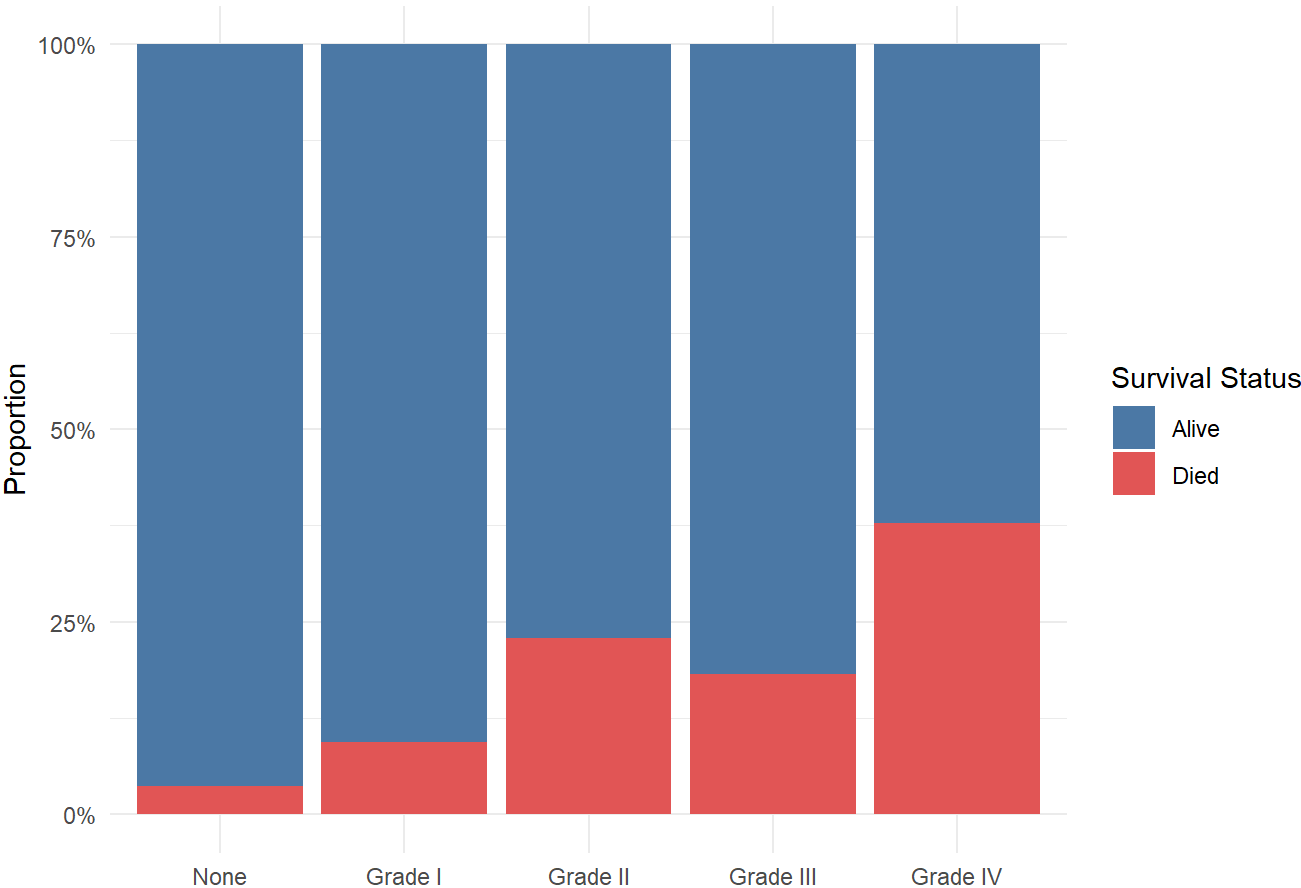
Distribution of Race/Ethnicity



Length of Stay by IVH Grade



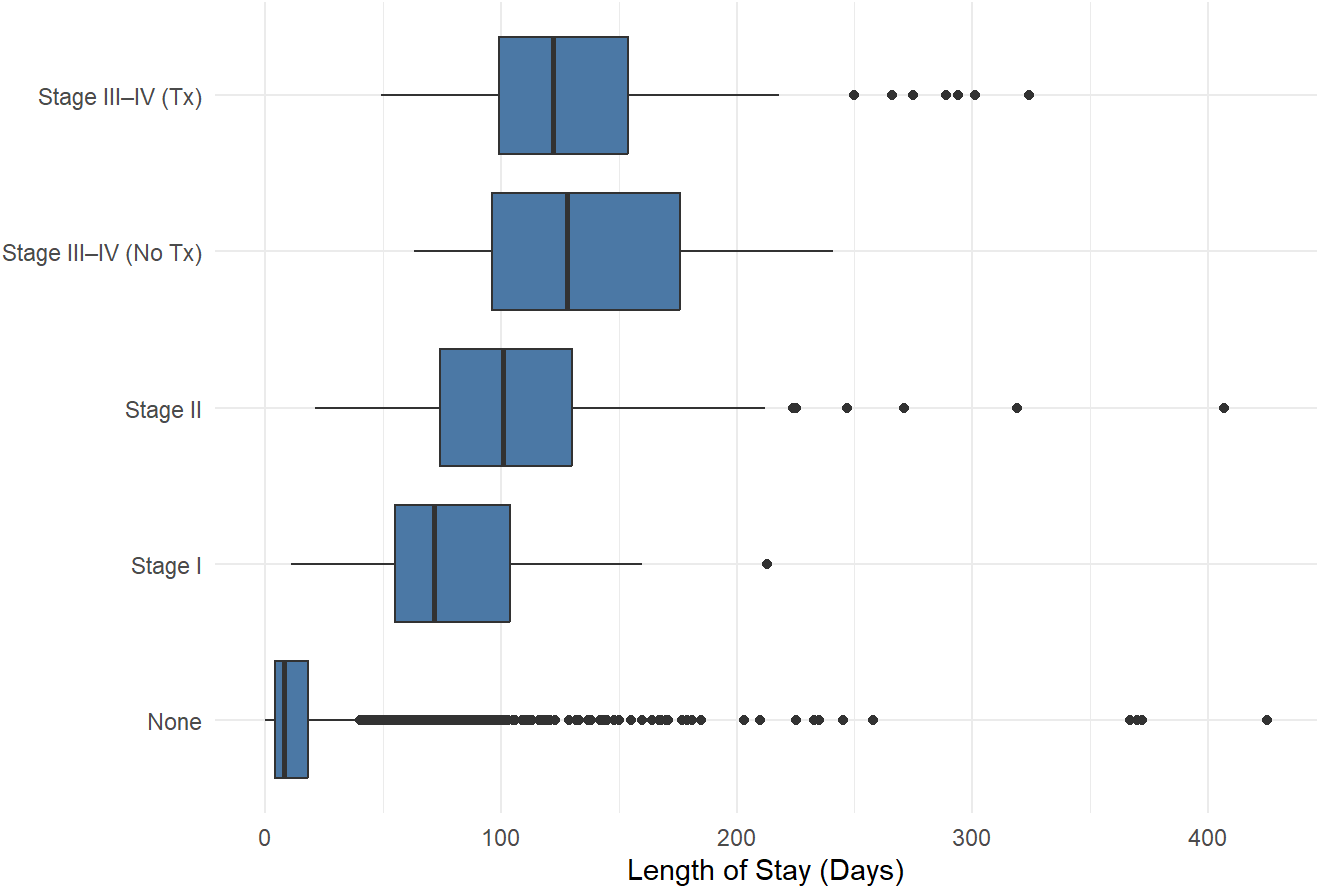
Survival Outcomes by IVH Grade



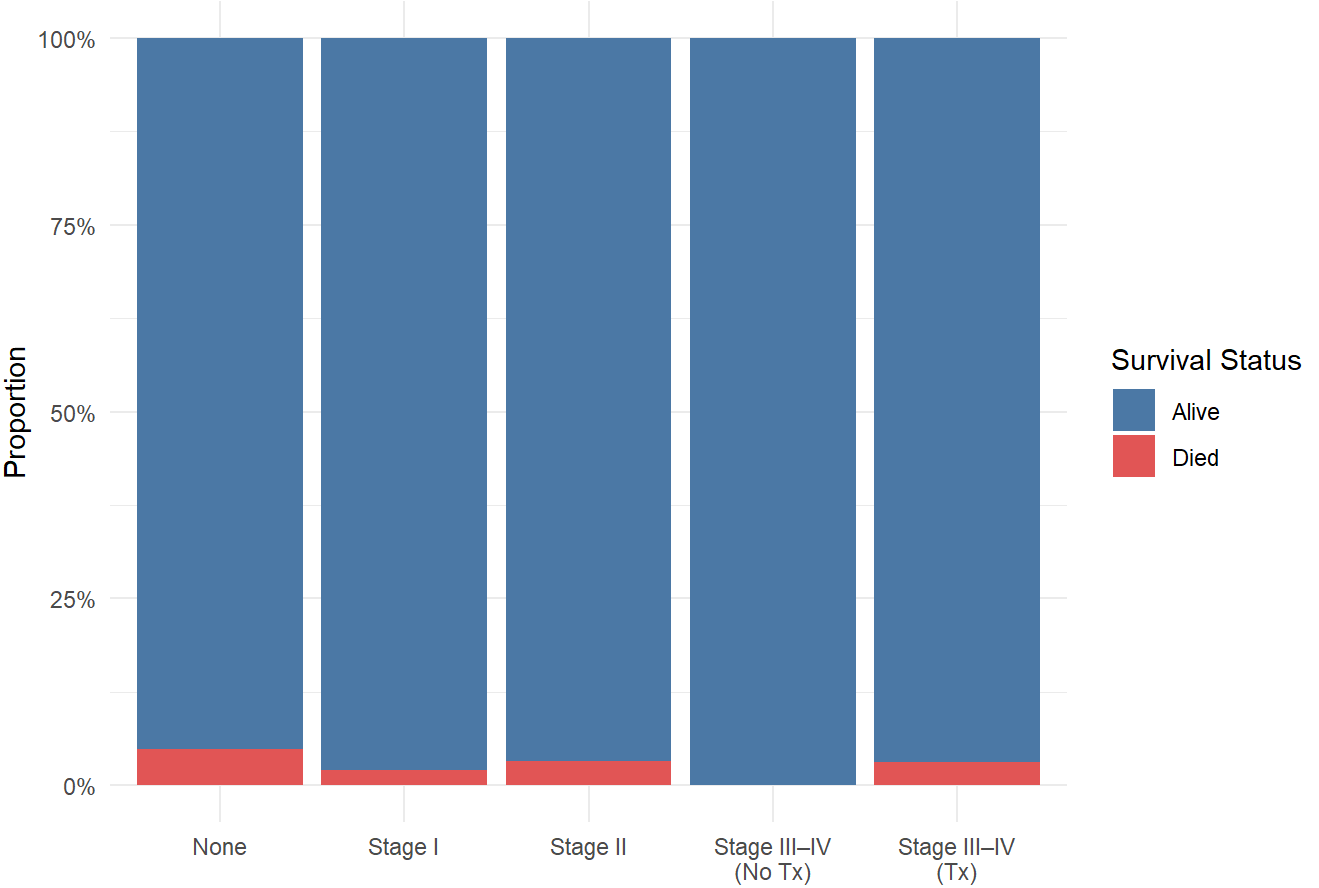
```
## # A tibble: 6 × 3
##   ivh          n percent
##   <fct>      <int>   <dbl>
## 1 Grade I      202    4.64
## 2 Grade II     48     1.1
## 3 Grade III    22     0.51
## 4 Grade IV     66     1.52
## 5 None      3964    91.1
## 6 Missing     48     1.1
```

```
## # A tibble: 6 × 7
##   ivh          n mean_los median_los IQR_los min_los max_los
##   <fct>      <int>   <dbl>      <dbl>  <dbl>  <dbl>  <dbl>
## 1 Grade I      202    56.8        43     65      0    324
## 2 Grade II     48    94.4       95.5   105.      5    301
## 3 Grade III    22    56.9        42    53.5     0    250
## 4 Grade IV     66    79.6       57.5   119.      1    275
## 5 None      3964    22.3         8     18     0    425
## 6 Missing     48    15.4        14    21.5     0     33
```

Length of Stay by ROP Stage

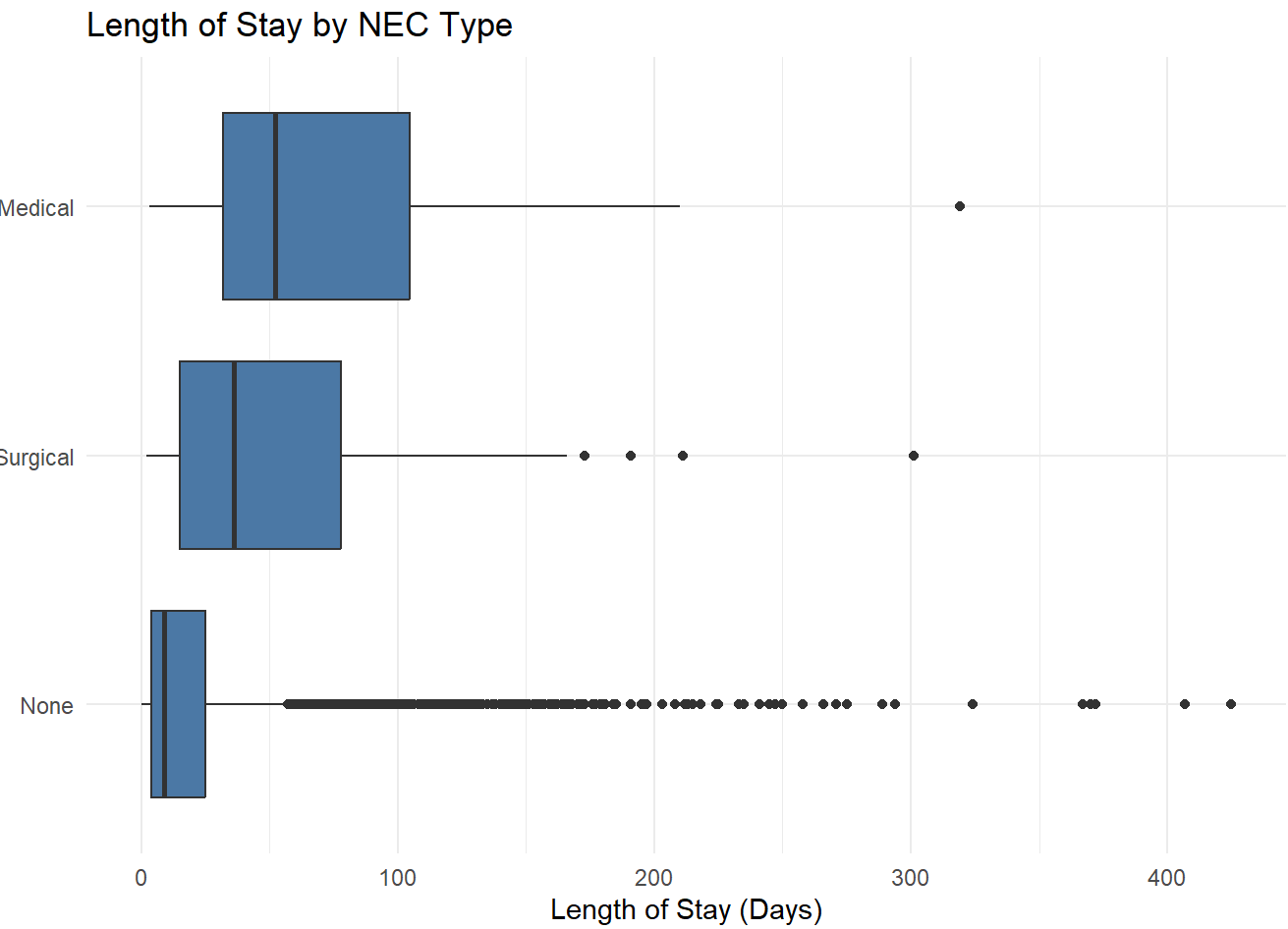


Survival Outcomes by ROP Stage

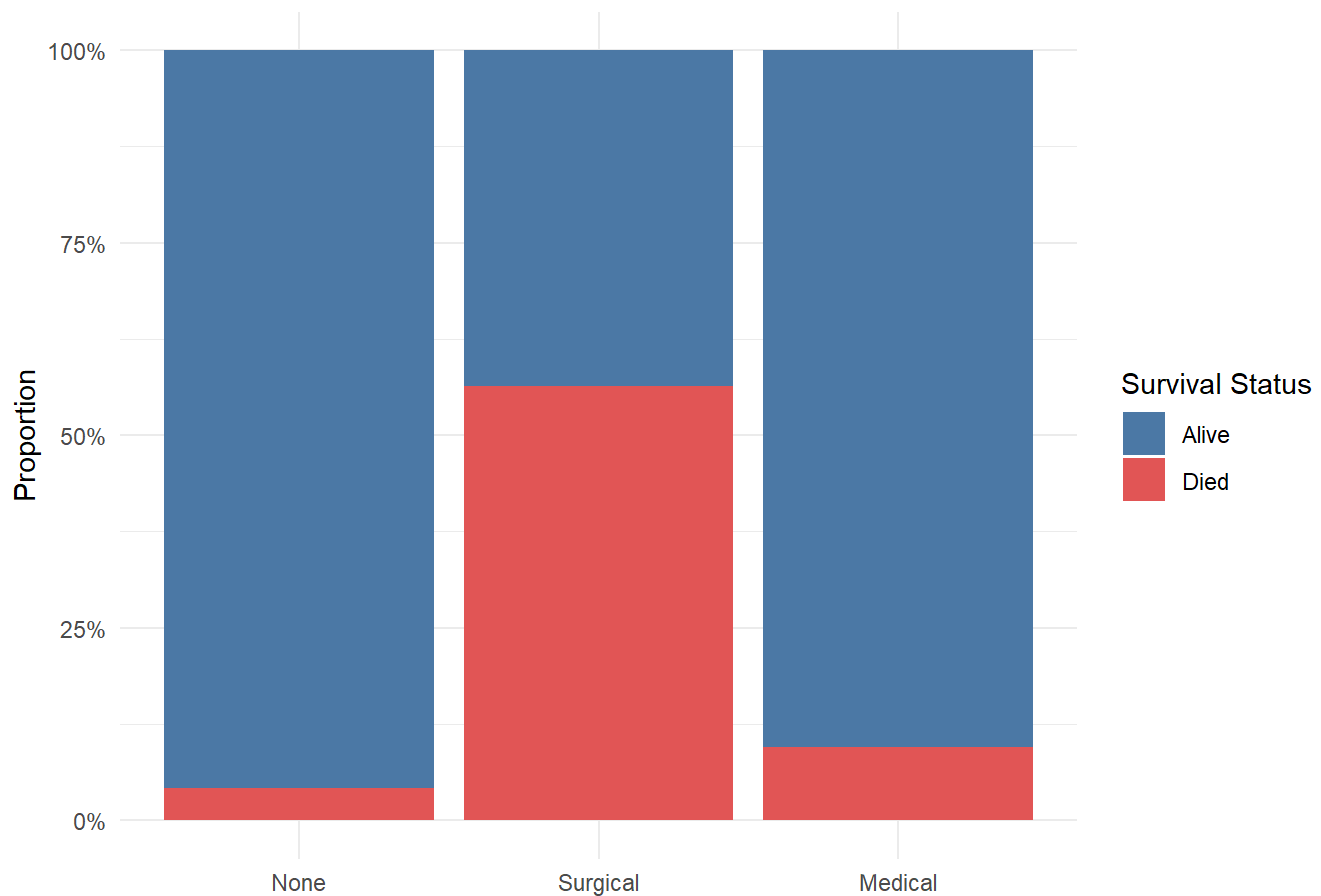


```
## # A tibble: 5 × 3
##   rop                                n percent
##   <fct>                        <int>   <dbl>
## 1 None                        3943    90.6
## 2 Stage I                     102     2.34
## 3 Stage II                     189     4.34
## 4 Stage III-IV (No Cryo or Laser Surgery)    19     0.44
## 5 Stage III-IV (With Cryo or Laser Surgery)   97     2.23
```

```
## # A tibble: 5 × 7
##   rop                                n mean_los median_los IQR_los min_los max_los
##   <fct>                        <int>   <dbl>     <dbl>   <dbl>   <dbl>   <dbl>
## 1 None                        3943    17.0         8      14         0     425
## 2 Stage I                     102    81.2       71.5     49        11     213
## 3 Stage II                     189   107.       101     56        21     407
## 4 Stage III-IV (No Cryo or La...    19   135       128     80        63     241
## 5 Stage III-IV (With Cryo or ...   97   135.       122     55        49     324
```



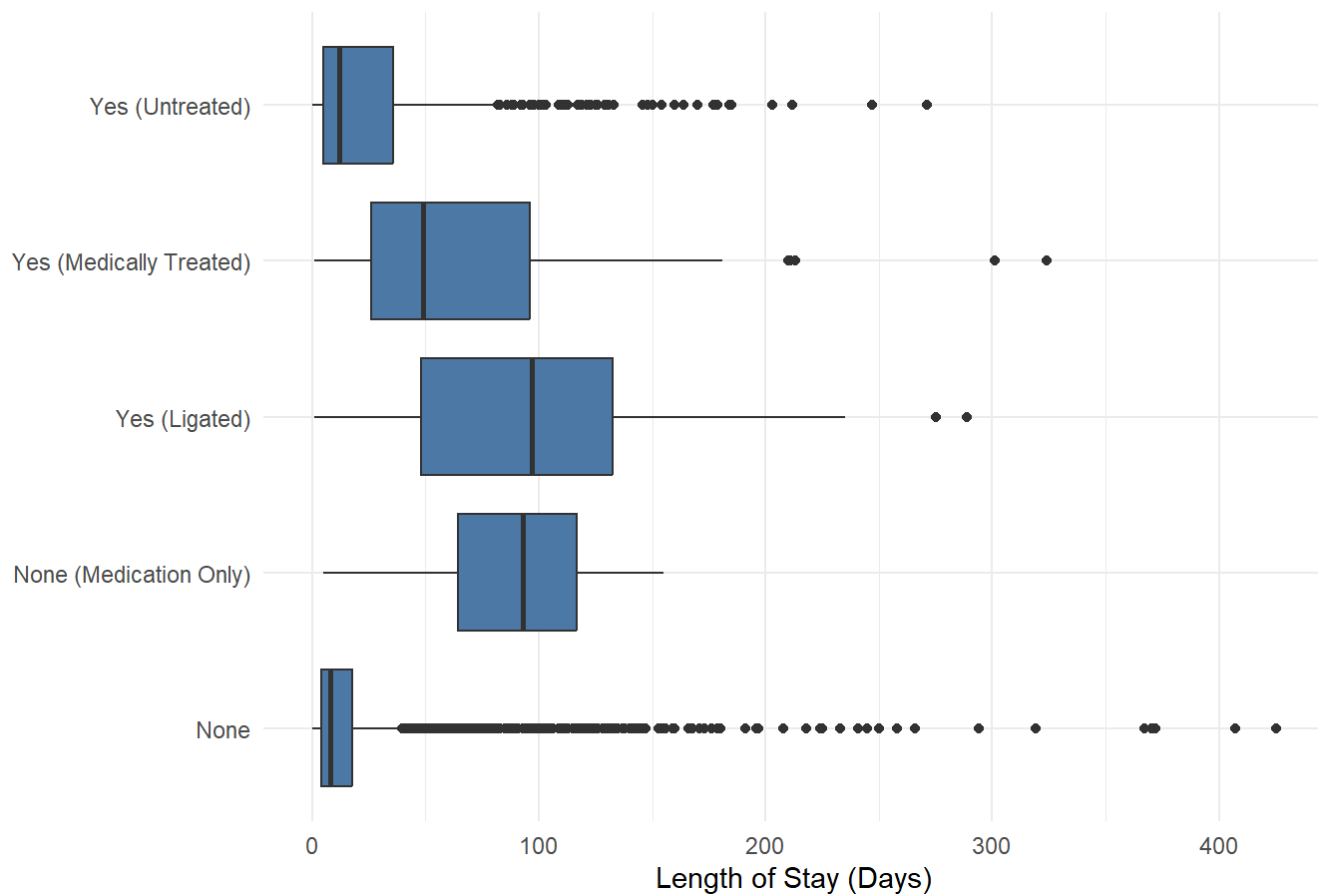
Survival Outcomes by NEC Type



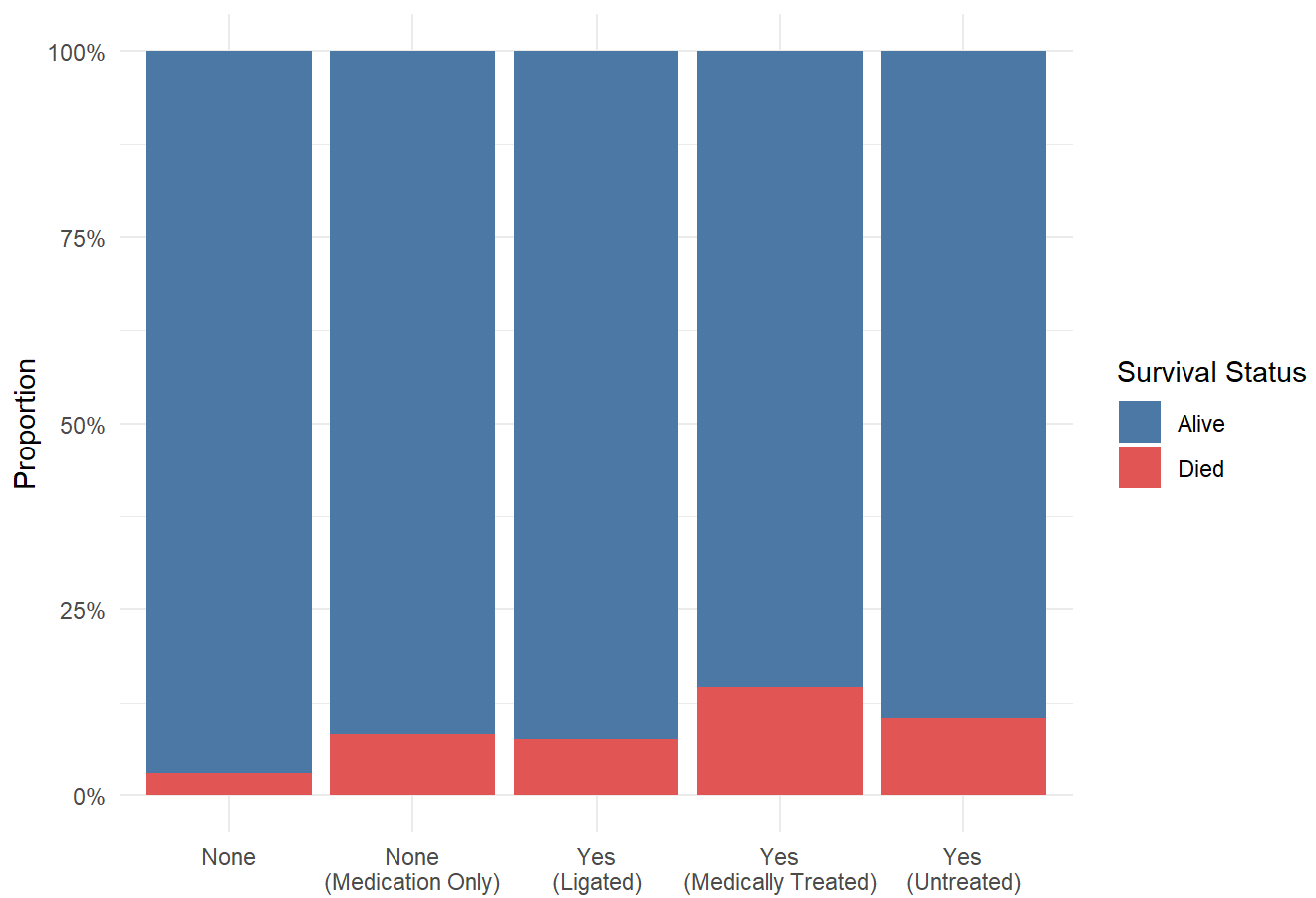
```
## # A tibble: 4 × 3
##   nec      n percent
##   <fct> <int>   <dbl>
## 1 Medical     63    1.45
## 2 None    4199   96.5
## 3 Surgical    39    0.9
## 4 Missing    49    1.13
```

```
## # A tibble: 4 × 7
##   nec      n mean_los median_los IQR_los min_los max_los
##   <fct> <int>   <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
## 1 Medical     63    71.1         52    72.5     3    319
## 2 None    4199    24.7          9    21      0    425
## 3 Surgical    39    67.9         36    63      2    301
## 4 Missing    49    15.4         14   21.5     0     33
```

Length of Stay by PDA Management

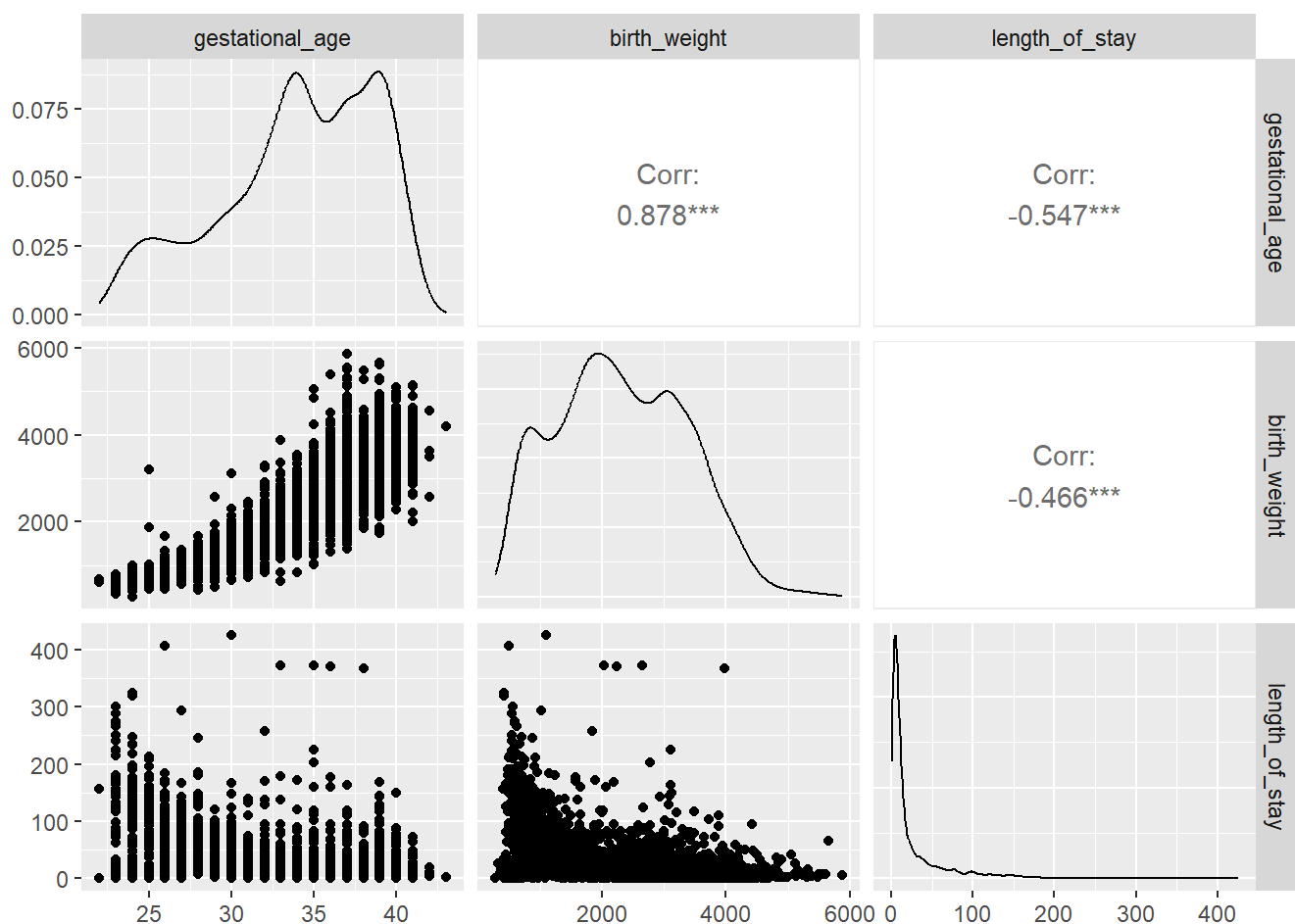


Survival Outcomes by PDA Management



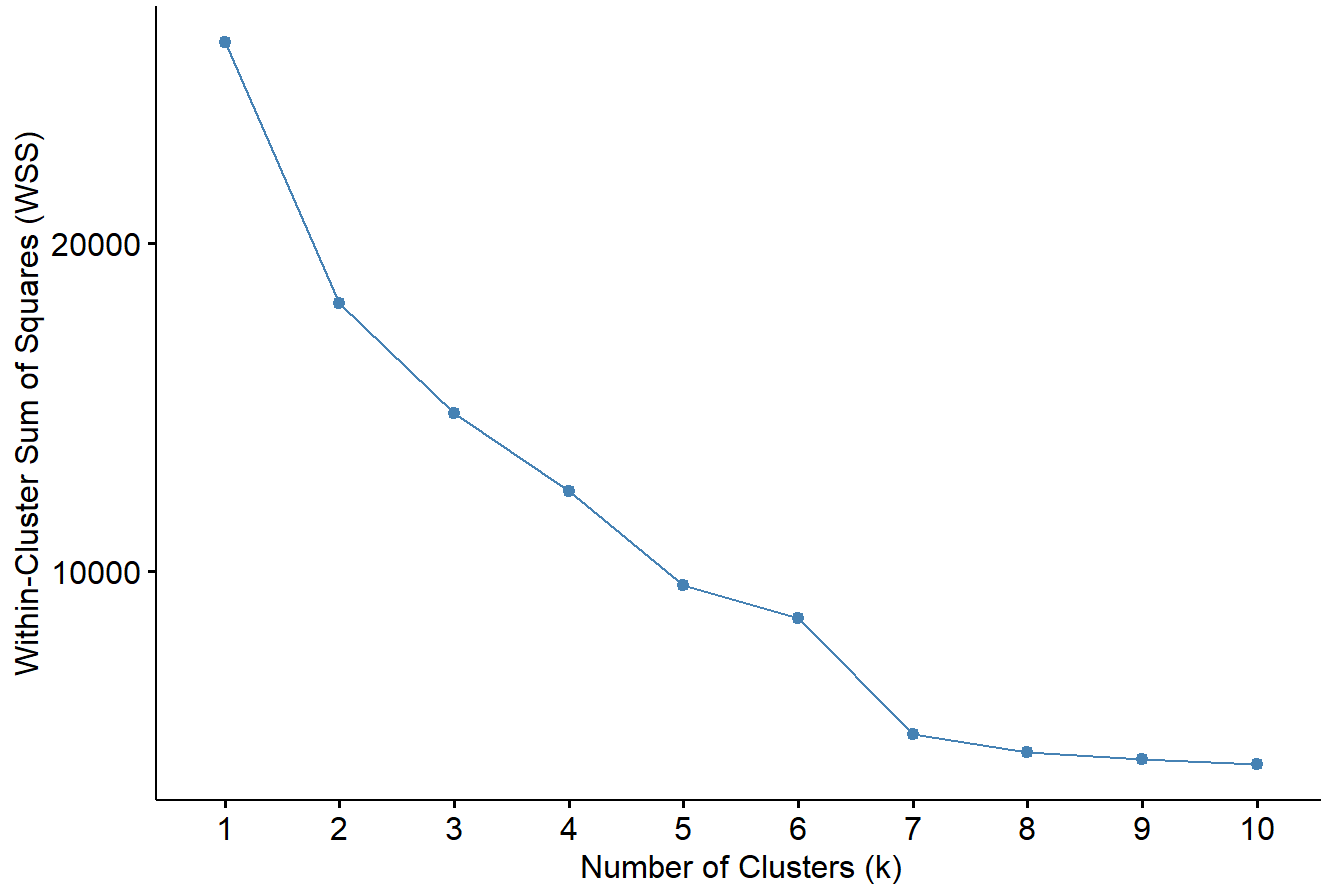
```
## # A tibble: 6 × 3
##   pda                n percent
##   <fct>            <int>   <dbl>
## 1 None             3358    77.2
## 2 None (Medication Only)    12    0.28
## 3 Yes (Ligated)           117    2.69
## 4 Yes (Medically Treated)   205    4.71
## 5 Yes (Untreated)          611   14.0
## 6 Missing              47    1.08
```

```
## # A tibble: 6 × 7
##   pda                n mean_los median_los IQR_los min_los max_los
##   <fct>            <int>   <dbl>     <dbl>   <dbl>   <dbl>   <dbl>
## 1 None             3358    20.3         8     14         0     425
## 2 None (Medication Only)    12    88.1        93    52.2         5     155
## 3 Yes (Ligated)           117    95.4        97    84.5         1     289
## 4 Yes (Medically Treated)   205    67.3        49    70         1     324
## 5 Yes (Untreated)          611    27.6        12    30.8         0     271
## 6 Missing              47    15.4        14    21.5         0      33
```



K-Means Clustering

Elbow Plot for Determining the Number of Clusters



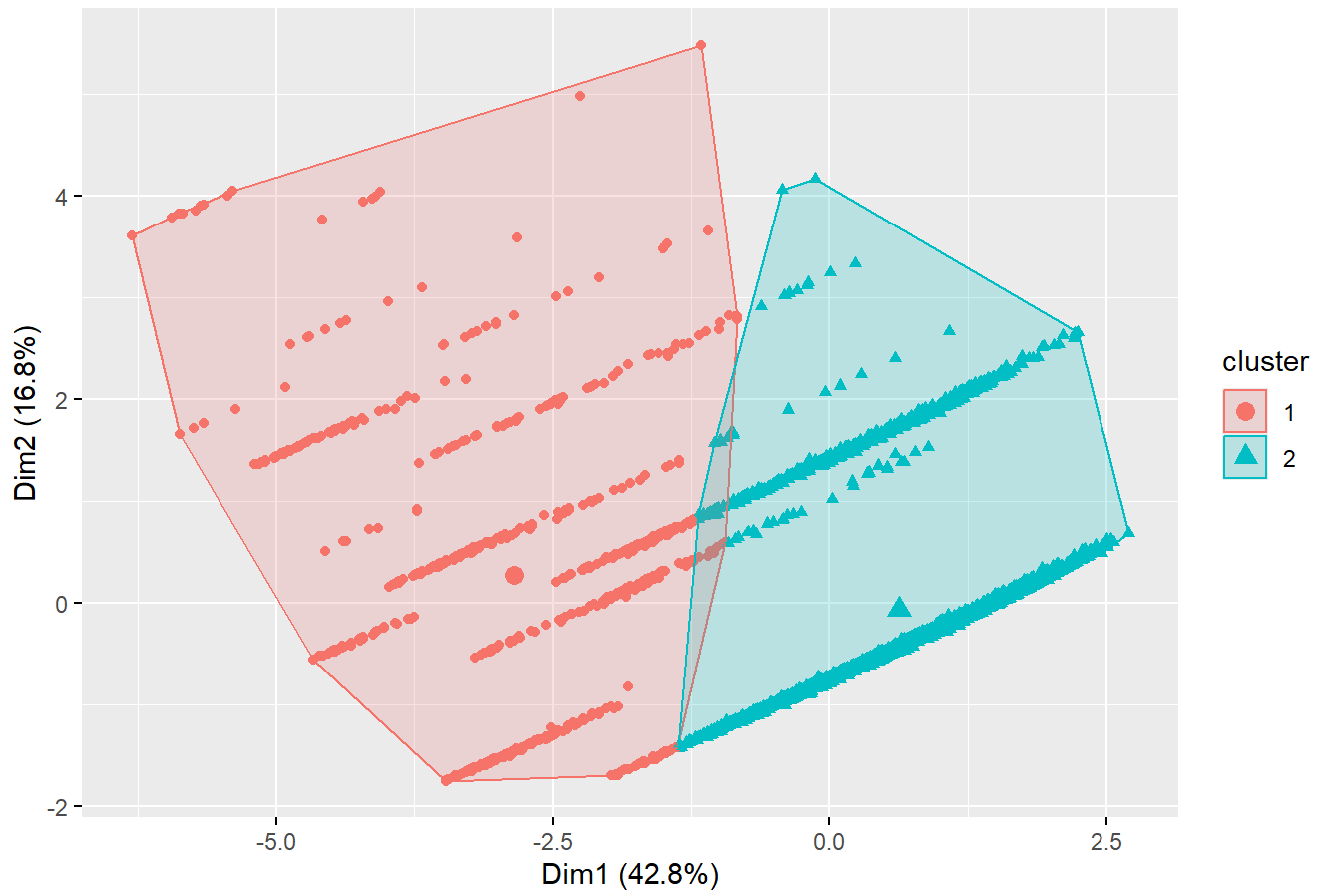
```
## # A tibble: 2 × 8
##   cluster_2 mean_ga mean_bw ivh_rate rop_rate nec_rate pda_rate mean_los
##   <fct>      <dbl>  <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 1          26.8    967.   0.383    0.518  0.109    0.496    77.0
## 2 2          35.6   2605.   0.0104    0      0.00449  0.152    14.5
```

```
## # A tibble: 4 × 8
##   cluster_4 mean_ga mean_bw ivh_rate rop_rate nec_rate pda_rate mean_los
##   <fct>      <dbl>  <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 1          28.8   1312.   0.255    0.294     1    0.422    69.9
## 2 2          38.2   3304.   0.0182    0       0    0.221    11.5
## 3 3          32.3   1786.   0.00106    0       0    0.137    19.9
## 4 4          26.9    994.   0.508    0.692     0    0.424    84.7
```

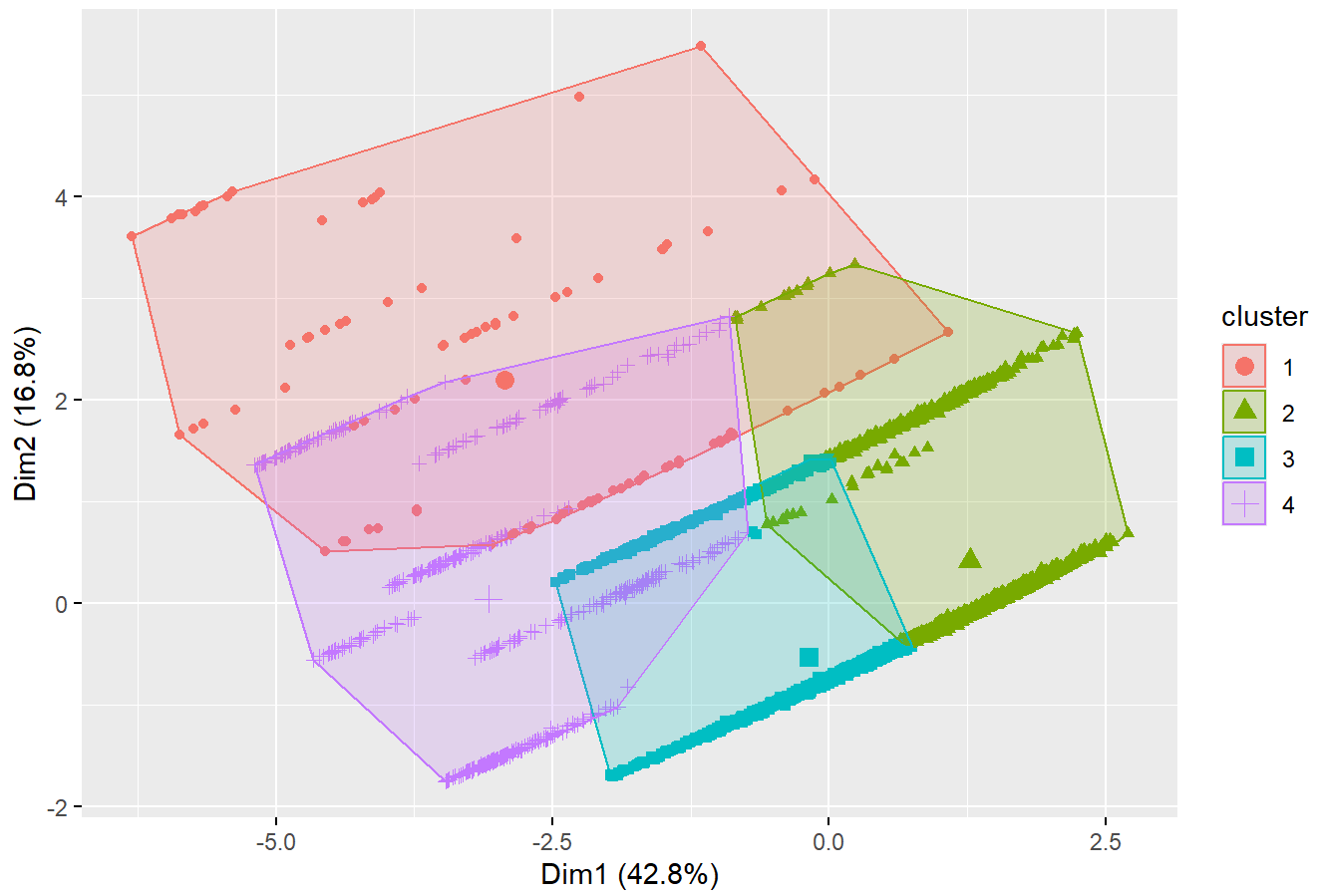
```
## # A tibble: 5 × 8
##   cluster_5 mean_ga mean_bw ivh_rate rop_rate nec_rate pda_rate mean_los
##   <fct>      <dbl>  <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 1          25.9    847.    0.289     1         0    0.459    107.
## 2 2          30.8   1660.     1         0         0    0.345     32.4
## 3 3          28.8   1312.    0.255    0.294         1    0.422     69.9
## 4 4          38.2   3304.     0         0         0    0.218     11.3
## 5 5          32.3   1786.     0         0         0    0.137     19.9
```

```
## # A tibble: 7 × 8
##   cluster_7 mean_ga mean_bw ivh_rate rop_rate nec_rate pda_rate mean_los
##   <fct>      <dbl>  <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 1          29.3   1268.     0         0         0     1       40.2
## 2 2          37.8   3206.  0.00236     0         0     1       16.8
## 3 3          38.3   3311.  0.00287     0         0     0        9.92
## 4 4          28.8   1312.    0.255    0.294         1    0.422     69.9
## 5 5          28.7   1321.     1    0.355         0    0.456     65.0
## 6 6          32.7   1849.     0         0         0     0       16.9
## 7 7          26.0    860.     0         1         0    0.381    100.
```

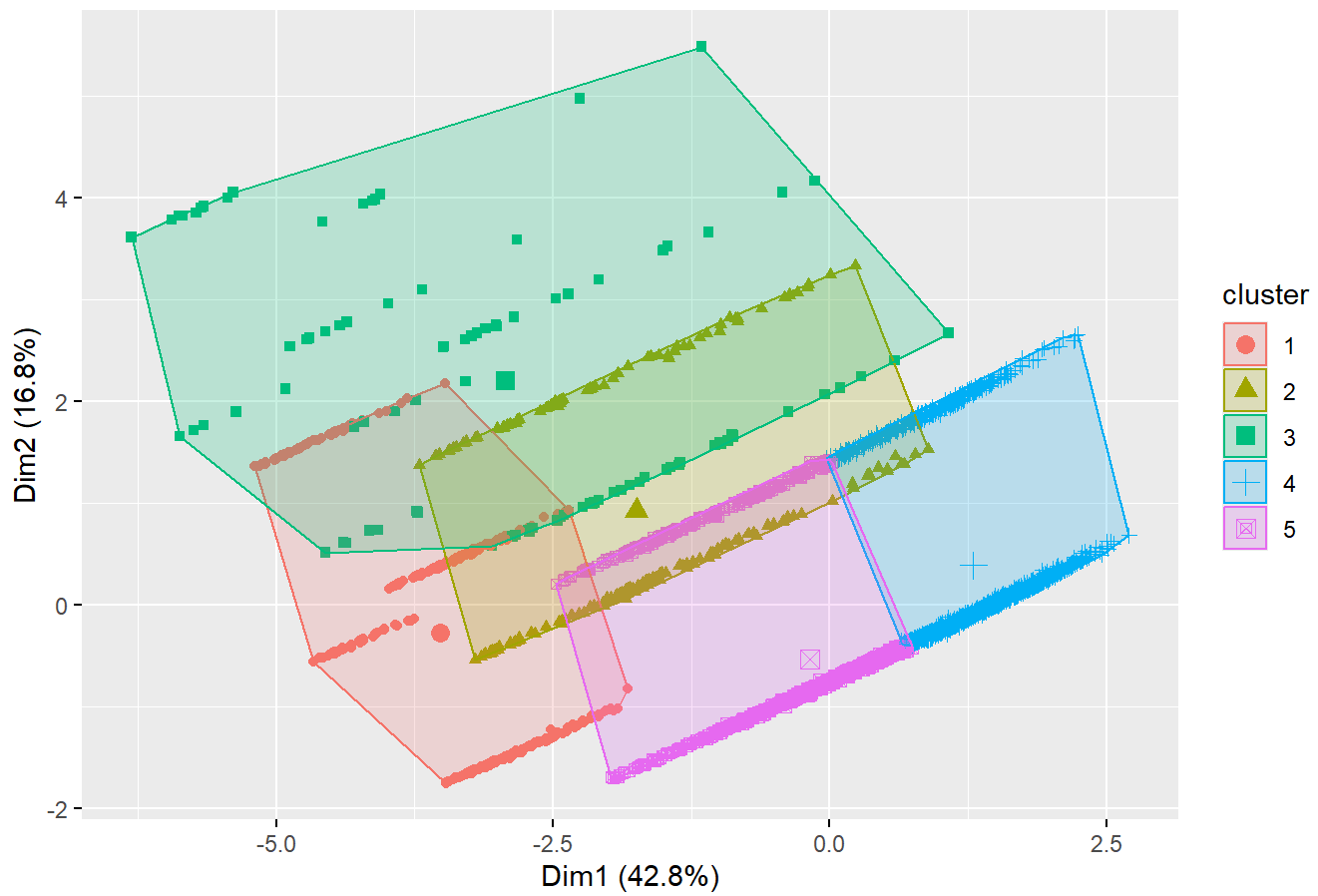
K-Means Clustering (k = 2)



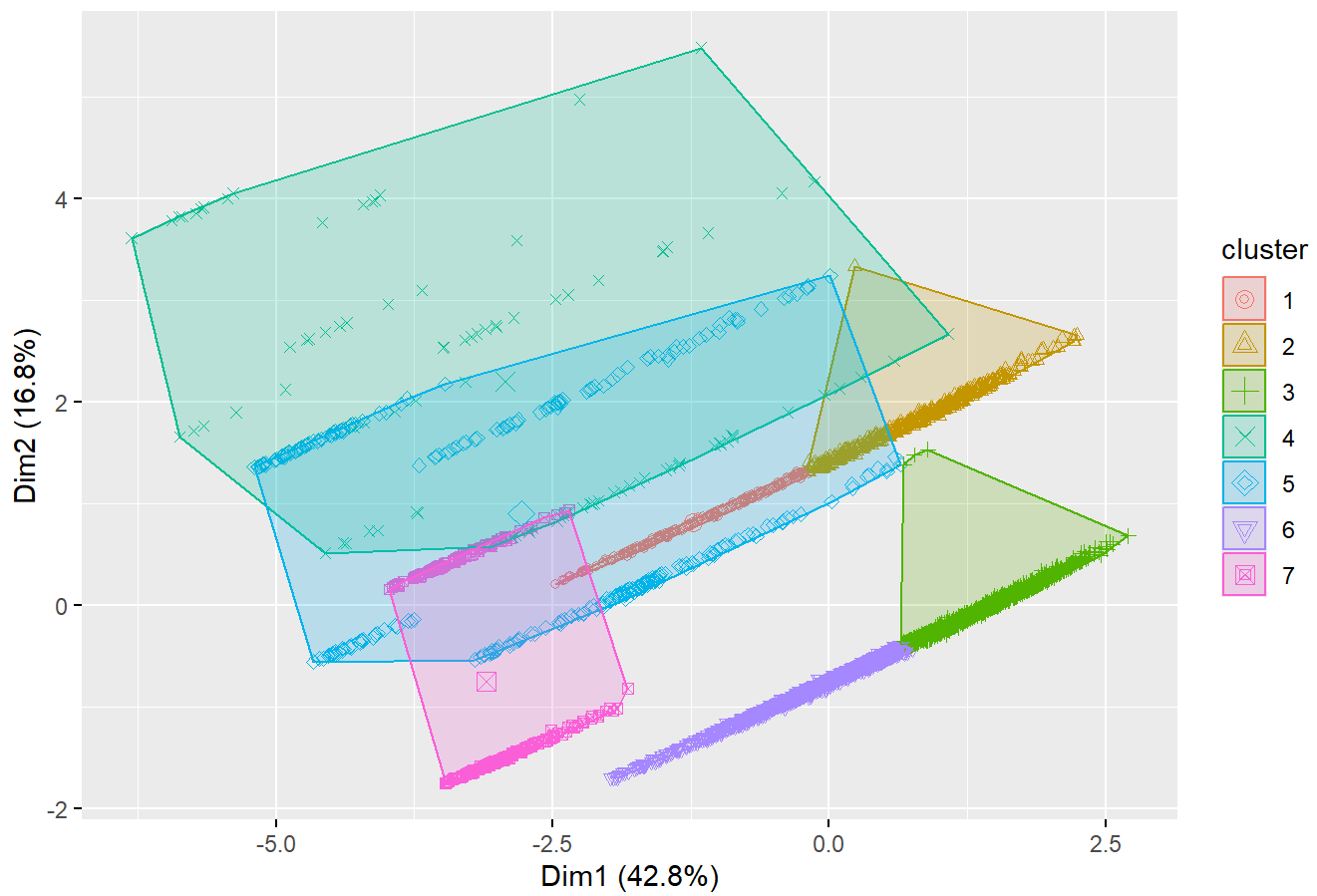
K-Means Clustering (k = 4)



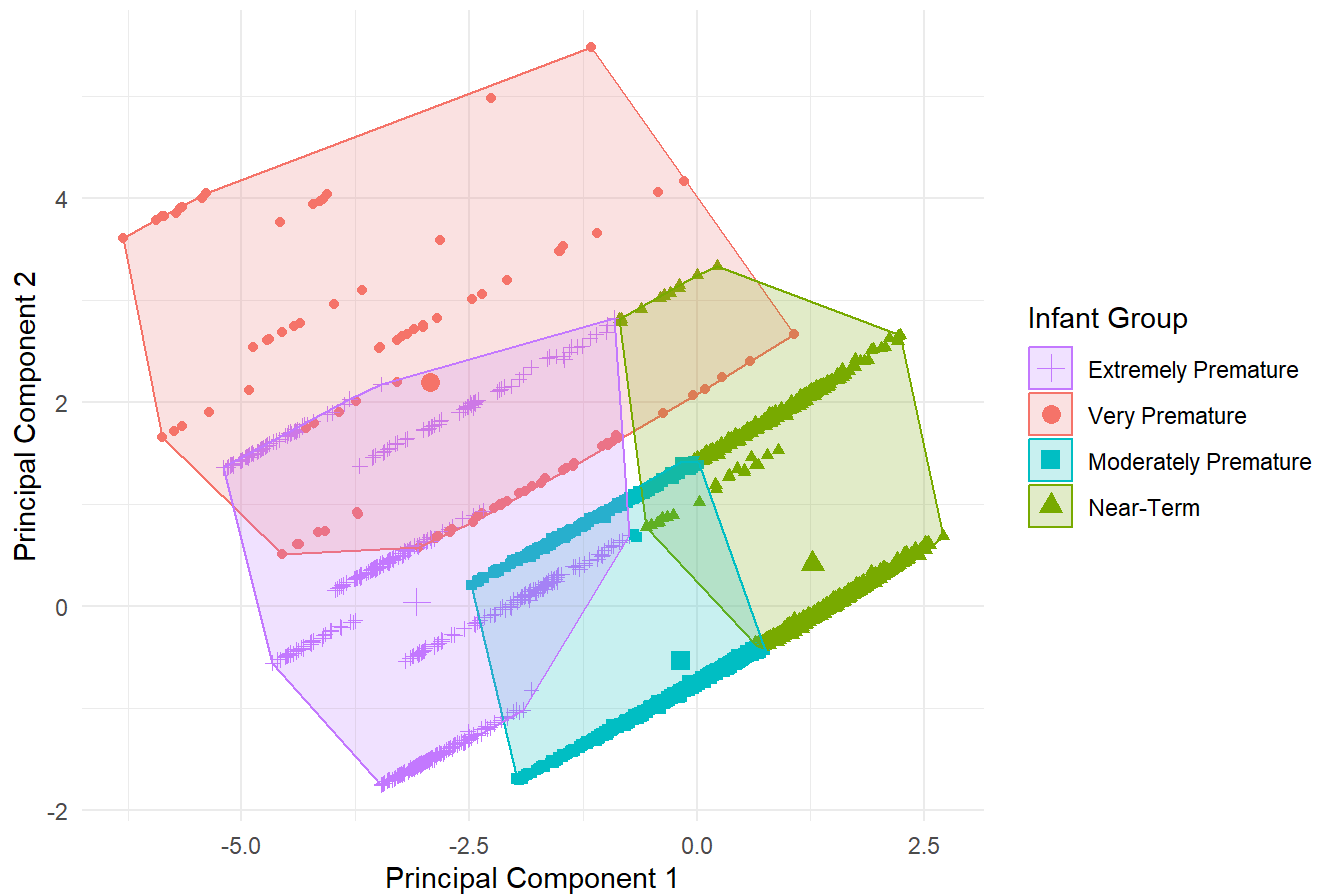
K-Means Clustering (k = 5)



K-Means Clustering (k = 7)

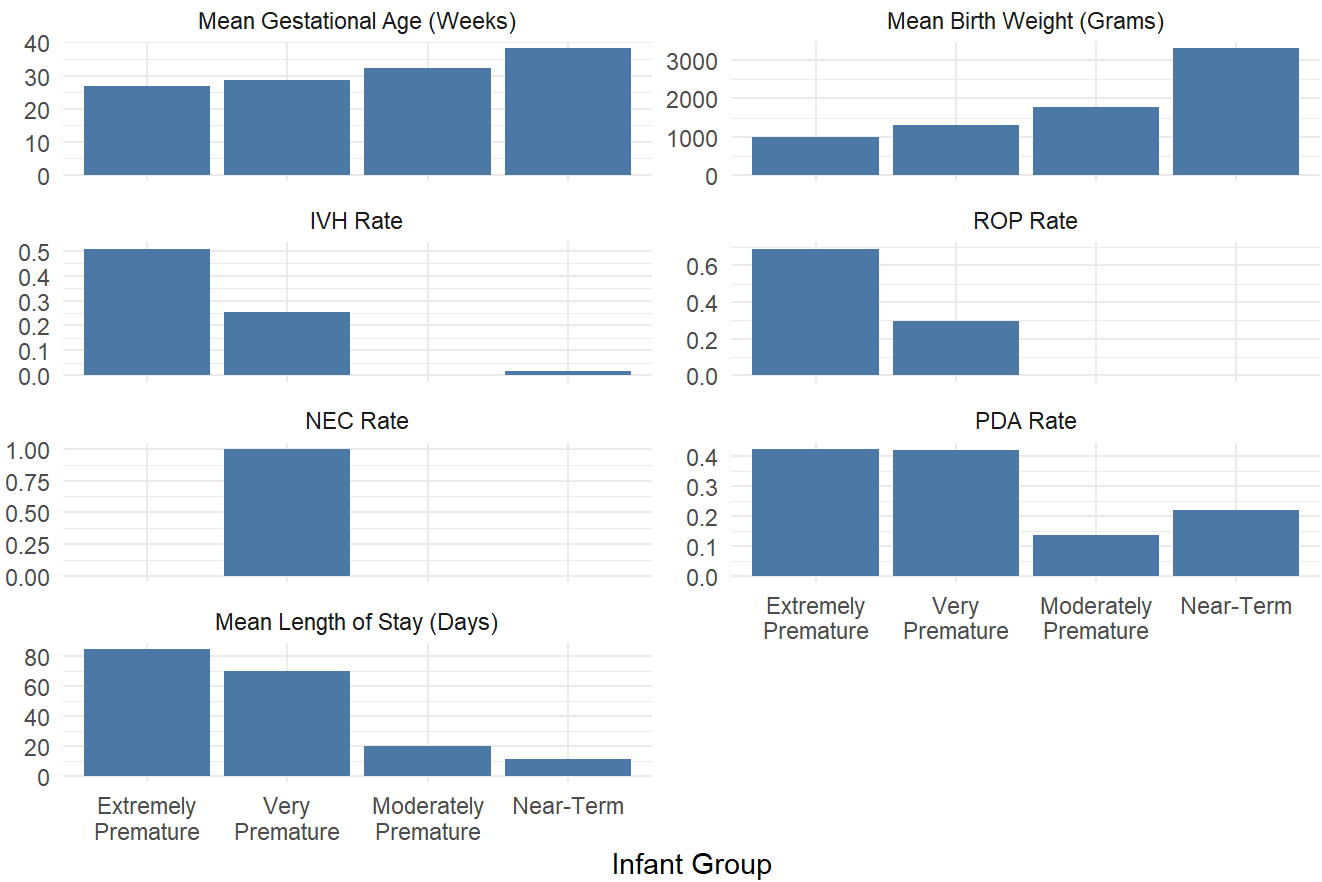


K-Means Clustering of Infant Groups (k = 4)



```
## # A tibble: 4 × 8
##   group_4      mean_ga mean_bw ivh_rate rop_rate nec_rate pda_rate mean_los
##   <fct>      <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 Extremely Premat... 26.9    994.   0.508   0.692     0    0.424    84.7
## 2 Very Premature    28.8   1312.   0.255   0.294     1    0.422    69.9
## 3 Moderately Prema... 32.3   1786.   0.00106 0         0    0.137    19.9
## 4 Near-Term         38.2   3304.   0.0182 0         0    0.221    11.5
```

Clinical Profiles of Infant Groups (k = 4)



Modeling

Initial Split

```
## Rows: 4,297
## Columns: 21
## $ patient_id      <dbl> 1384, 4010, 2751, 1035, 1117, 2180, 2608, 603, 435, 16...
## $ survival_status <fct> Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive...
## $ gestational_age  <dbl> 34, 30, 30, 33, 27, 26, 38, 32, 32, 30, 30, 38, 31, 35...
## $ birth_weight     <dbl> 2700, 1250, 1380, 2190, 910, 830, 2850, 1900, 1520, 12...
## $ gender           <fct> Male, Female, Male, Female, Male, Male, Female, Female...
## $ inborn           <fct> Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes,...
## $ race_ethnicity   <fct> Other, White, White, White, White, Black, White, Black...
## $ length_of_stay   <dbl> 15, 5, 5, 3, 52, 46, 15, 5, 5, 52, 52, 15, 33, 10, 3, ...
## $ ivh              <fct> None, None, None, None, None, None, None, None, None, ...
## $ rop              <fct> None, None, None, None, Stage I, None, None, None, Non...
## $ nec              <fct> None, None, None, None, Medical, None, None, None, Non...
## $ pda              <fct> None, None (Medication Only), None, None, Yes (Ligated...
## $ ivh_2            <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ rop_2            <fct> No, No, No, No, Yes, No, No, No, No, Yes, No, No, No, ...
## $ nec_2            <fct> No, No, No, No, Yes, No, No, No, No, No, No, No, No, N...
## $ pda_2            <fct> No, No, No, No, Yes, Yes, No, No, No, Yes, Yes, No, Ye...
## $ cluster_2        <fct> 2, 2, 2, 2, 1, 1, 2, 2, 2, 1, 1, 2, 2, 2, 2, 2, 2, ...
## $ cluster_4        <fct> 3, 3, 3, 3, 1, 3, 2, 3, 3, 4, 3, 2, 3, 3, 3, 3, 2, 2, ...
## $ cluster_5        <fct> 5, 5, 5, 5, 3, 5, 4, 5, 5, 1, 5, 4, 5, 5, 5, 5, 4, 4, ...
## $ cluster_7        <fct> 6, 6, 6, 6, 4, 1, 3, 6, 6, 7, 1, 3, 1, 6, 6, 6, 3, 3, ...
## $ group_4          <fct> Moderately Premature, Moderately Premature, Moderately...
```

```

## Rows: 3,437
## Columns: 21
## $ patient_id      <dbl> 2797, 302, 2794, 1376, 463, 555, 1941, 2503, 357, 3152...
## $ survival_status <fct> Alive, Alive, Died, Alive, Alive, Alive, Alive, Alive,...
## $ gestational_age <dbl> 27, 34, 38, 34, 37, 34, 40, 35, 32, 38, 32, 37, 30, 31...
## $ birth_weight    <dbl> 1180, 2125, 3180, 2280, 2990, 2460, 3620, 3010, 1380, ...
## $ gender          <fct> Male, Female, Female, Male, Female, Female, Male, Male...
## $ inborn          <fct> Yes, Yes, Yes, No, Yes, Yes, Yes, No, Yes, Yes, Yes, Y...
## $ race_ethnicity  <fct> White, White, Other, White, White, Black, Black, White...
## $ length_of_stay  <dbl> 106, 8, 1, 15, 29, 3, 2, 102, 8, 32, 25, 2, 51, 30, 10...
## $ ivh             <fct> Grade I, None, None, None, None, None, None, None, Non...
## $ rop             <fct> Stage III-IV (With Cryo or Laser Surgery), None, None,...
## $ nec             <fct> None, None, None, None, None, None, None, None, None, ...
## $ pda             <fct> Yes (Medically Treated), None, Yes (Untreated), None, ...
## $ ivh_2           <fct> Yes, No, No, No, No, No, No, No, No, No, No, No, Yes, ...
## $ rop_2           <fct> Yes, No, No, No, No, No, No, No, No, No, No, No, No, N...
## $ nec_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ pda_2           <fct> Yes, No, Yes, No, Yes, Yes, No, No, No, Yes, No, No, N...
## $ cluster_2       <fct> 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 1, ...
## $ cluster_4       <fct> 4, 3, 2, 3, 2, 3, 2, 2, 3, 2, 3, 2, 4, 3, 2, 3, 2, 3, ...
## $ cluster_5       <fct> 1, 5, 4, 5, 4, 5, 4, 4, 5, 4, 5, 4, 2, 5, 4, 5, 4, 5, ...
## $ cluster_7       <fct> 5, 6, 2, 6, 2, 2, 3, 3, 6, 2, 6, 3, 5, 6, 3, 6, 2, 6, ...
## $ group_4         <fct> Extremely Premature, Moderately Premature, Near-Term, ...

```

```

## Rows: 860
## Columns: 21
## $ patient_id      <dbl> 3901, 3584, 109, 1074, 408, 2743, 4345, 1547, 1622, 43...
## $ survival_status <fct> Alive, Alive, Alive, Alive, Alive, Alive, Died, Alive,...
## $ gestational_age <dbl> 26, 36, 33, 34, 37, 40, 24, 40, 34, 31, 38, 33, 28, 36...
## $ birth_weight    <dbl> 500, 4110, 2180, 2095, 2635, 3315, 659, 3140, 2560, 20...
## $ gender          <fct> Female, Male, Male, Male, Female, Female, Male, Male, ...
## $ inborn          <fct> Yes, Yes, Yes, Yes, No, Yes, No, Yes, Yes, Yes, Yes, Y...
## $ race_ethnicity  <fct> White, Black, Other, White, Black, Other, Black, White...
## $ length_of_stay  <dbl> 124, 3, 33, 7, 7, 2, 129, 1, 21, 6, 6, 14, 34, 5, 2, 7...
## $ ivh             <fct> Grade I, None, None, None, None, None, None, None, Non...
## $ rop             <fct> Stage II, None, None, None, None, None, None, None, No...
## $ nec             <fct> None, None, None, None, None, None, None, None, None, ...
## $ pda             <fct> None, None, Yes (Medically Treated), None, None, None,...
## $ ivh_2           <fct> Yes, No, No, No, No, No, No, No, No, No, No, No, No, N...
## $ rop_2           <fct> Yes, No, No, No, No, No, No, No, No, No, No, No, No, N...
## $ nec_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, Yes, N...
## $ pda_2           <fct> No, No, Yes, No, No, No, No, No, No, No, No, No, No, N...
## $ cluster_2       <fct> 1, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 1, 2, 2, 1, 2, 2, ...
## $ cluster_4       <fct> 4, 2, 3, 3, 2, 2, 3, 2, 3, 3, 2, 3, 1, 2, 2, 4, 3, 3, ...
## $ cluster_5       <fct> 1, 4, 5, 5, 4, 4, 5, 4, 5, 5, 4, 5, 3, 4, 4, 1, 5, 5, ...
## $ cluster_7       <fct> 5, 3, 1, 6, 3, 3, 6, 3, 6, 6, 3, 6, 4, 3, 3, 7, 6, 6, ...
## $ group_4         <fct> Extremely Premature, Near-Term, Moderately Premature, ...

```

Full Linear Regression Model

```
##
## Call:
## lm(formula = predictor_formula, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -92.00 -10.98  -4.51   3.52  390.51
##
## Coefficients: (2 not defined because of singularities)
##
##              Estimate Std. Error t value
## (Intercept)      1.107e+02  7.825e+00  14.142
## survival_statusDied -1.459e+01  2.442e+00  -5.975
## gestational_age    -1.697e+00  2.369e-01  -7.163
## birth_weight      -1.966e-03  9.515e-04  -2.066
## genderMale        -3.587e-01  9.441e-01  -0.380
## inbornYes         -1.106e+01  1.114e+00  -9.927
## race_ethnicityOther -8.386e-01  1.374e+00  -0.610
## race_ethnicityWhite -4.443e-02  1.019e+00  -0.044
## ivhGrade II        1.149e+01  5.062e+00   2.270
## ivhGrade III       -1.274e+01  7.611e+00  -1.674
## ivhGrade IV         3.951e+00  4.437e+00   0.890
## ivhNone            -9.051e+00  2.277e+00  -3.975
## ivhMissing         -2.086e+01  1.195e+01  -1.745
## ropStage I         4.332e+01  3.288e+00  13.178
## ropStage II        6.299e+01  2.631e+00  23.944
## ropStage III-IV (No Cryo or Laser Surgery) 8.889e+01  6.950e+00  12.789
## ropStage III-IV (With Cryo or Laser Surgery) 8.949e+01  3.588e+00  24.942
## necNone            -1.368e+01  3.911e+00  -3.499
## necSurgical        -6.107e+00  6.300e+00  -0.969
## necMissing          NA          NA      NA
## pdaNone (Medication Only) 8.718e+00  8.010e+00   1.088
## pdaYes (Ligated)      2.150e+01  3.164e+00   6.795
## pdaYes (Medically Treated) 7.335e+00  2.382e+00   3.079
## pdaYes (Untreated)    7.972e+00  1.380e+00   5.778
## pdaMissing          NA          NA      NA
##
##              Pr(>|t|)
## (Intercept)      < 2e-16 ***
## survival_statusDied 2.54e-09 ***
## gestational_age    9.60e-13 ***
## birth_weight      0.038862 *
## genderMale        0.704008
## inbornYes         < 2e-16 ***
## race_ethnicityOther 0.541700
## race_ethnicityWhite 0.965245
## ivhGrade II       0.023248 *
## ivhGrade III      0.094134 .
```

```

## ivhGrade IV 0.373269
## ivhNone 7.19e-05 ***
## ivhMissing 0.081106 .
## ropStage I < 2e-16 ***
## ropStage II < 2e-16 ***
## ropStage III-IV (No Cryo or Laser Surgery) < 2e-16 ***
## ropStage III-IV (With Cryo or Laser Surgery) < 2e-16 ***
## necNone 0.000473 ***
## necSurgical 0.332469
## necMissing NA
## pdaNone (Medication Only) 0.276514
## pdaYes (Ligated) 1.27e-11 ***
## pdaYes (Medically Treated) 0.002092 **
## pdaYes (Untreated) 8.25e-09 ***
## pdaMissing NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.14 on 3414 degrees of freedom
## Multiple R-squared:  0.5482, Adjusted R-squared:  0.5453
## F-statistic: 188.3 on 22 and 3414 DF,  p-value: < 2.2e-16

```

Log-Linear Regression Model

```
##
## Call:
## lm(formula = predictor_formula_log, data = train_data_2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4808 -0.5175 -0.0462  0.4551  3.8418
##
## Coefficients: (2 not defined because of singularities)
##
##              Estimate Std. Error t value
## (Intercept)      6.4479647   0.2385317   27.032
## survival_statusDied -0.9442598   0.0744492  -12.683
## gestational_age    -0.0677188   0.0072222   -9.377
## birth_weight     -0.0001949   0.0000290   -6.719
## genderMale         0.0040365   0.0287785    0.140
## inbornYes         -0.4735488   0.0339630  -13.943
## race_ethnicityOther  0.0287727   0.0418826    0.687
## race_ethnicityWhite  0.0871706   0.0310765    2.805
## ivhGrade II       -0.0489791   0.1543024   -0.317
## ivhGrade III      -0.2921972   0.2319901   -1.260
## ivhGrade IV       -0.2077987   0.1352371   -1.537
## ivhNone           -0.4088817   0.0694122   -5.891
## ivhMissing        -1.0784805   0.3644105   -2.960
## ropStage I         0.9433098   0.1002171    9.413
## ropStage II        1.0808605   0.0801862   13.479
## ropStage III-IV (No Cryo or Laser Surgery) 1.2245028   0.2118619    5.780
## ropStage III-IV (With Cryo or Laser Surgery) 1.1290542   0.1093612   10.324
## necNone           -0.6167862   0.1192149   -5.174
## necSurgical        -0.2486079   0.1920470   -1.295
## necMissing                NA                NA                NA
## pdaNone (Medication Only)  0.3105854   0.2441654    1.272
## pdaYes (Ligated)         0.4992957   0.0964497    5.177
## pdaYes (Medically Treated) 0.3900356   0.0726080    5.372
## pdaYes (Untreated)       0.4059981   0.0420572    9.653
## pdaMissing                NA                NA                NA
##
##              Pr(>|t|)
## (Intercept)      < 2e-16 ***
## survival_statusDied < 2e-16 ***
## gestational_age    < 2e-16 ***
## birth_weight     2.14e-11 ***
## genderMale        0.88846
## inbornYes         < 2e-16 ***
## race_ethnicityOther 0.49214
## race_ethnicityWhite 0.00506 **
## ivhGrade II        0.75094
## ivhGrade III       0.20793
```

```

## ivhGrade IV                0.12450
## ivhNone                    4.22e-09 ***
## ivhMissing                  0.00310 **
## ropStage I                  < 2e-16 ***
## ropStage II                 < 2e-16 ***
## ropStage III-IV (No Cryo or Laser Surgery) 8.15e-09 ***
## ropStage III-IV (With Cryo or Laser Surgery) < 2e-16 ***
## necNone                     2.43e-07 ***
## necSurgical                 0.19557
## necMissing                  NA
## pdaNone (Medication Only)   0.20345
## pdaYes (Ligated)            2.39e-07 ***
## pdaYes (Medically Treated)  8.32e-08 ***
## pdaYes (Untreated)          < 2e-16 ***
## pdaMissing                  NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8274 on 3414 degrees of freedom
## Multiple R-squared:  0.5028, Adjusted R-squared:  0.4996
## F-statistic: 156.9 on 22 and 3414 DF,  p-value: < 2.2e-16

```

Log-Linear Regression Model (Categorical Morbidities)

```
##
## Call:
## lm(formula = predictor_formula_log_binary, data = train_data_2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4304 -0.5187 -0.0471  0.4629  3.8408
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.453e+00  2.017e-01  27.035  < 2e-16 ***
## survival_statusDied -9.859e-01  7.200e-02 -13.694  < 2e-16 ***
## gestational_age    -6.846e-02  7.097e-03  -9.647  < 2e-16 ***
## birth_weight     -1.954e-04  2.892e-05  -6.757  1.65e-11 ***
## genderMale        2.874e-03  2.872e-02   0.100  0.92030
## inbornYes        -4.758e-01  3.379e-02 -14.082  < 2e-16 ***
## race_ethnicityOther 3.195e-02  4.178e-02   0.765  0.44448
## race_ethnicityWhite 8.794e-02  3.103e-02   2.834  0.00463 **
## ivh_2Yes          3.518e-01  5.762e-02   6.106  1.13e-09 ***
## rop_2Yes          1.063e+00  6.226e-02  17.070  < 2e-16 ***
## nec_2Yes          5.482e-01  9.529e-02   5.753  9.56e-09 ***
## pda_2Yes          4.079e-01  3.618e-02  11.277  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8274 on 3425 degrees of freedom
## Multiple R-squared:  0.5013, Adjusted R-squared:  0.4997
## F-statistic: 312.9 on 11 and 3425 DF,  p-value: < 2.2e-16
```

Cross-Validation for Full, Log, and Categorical Linear Regression

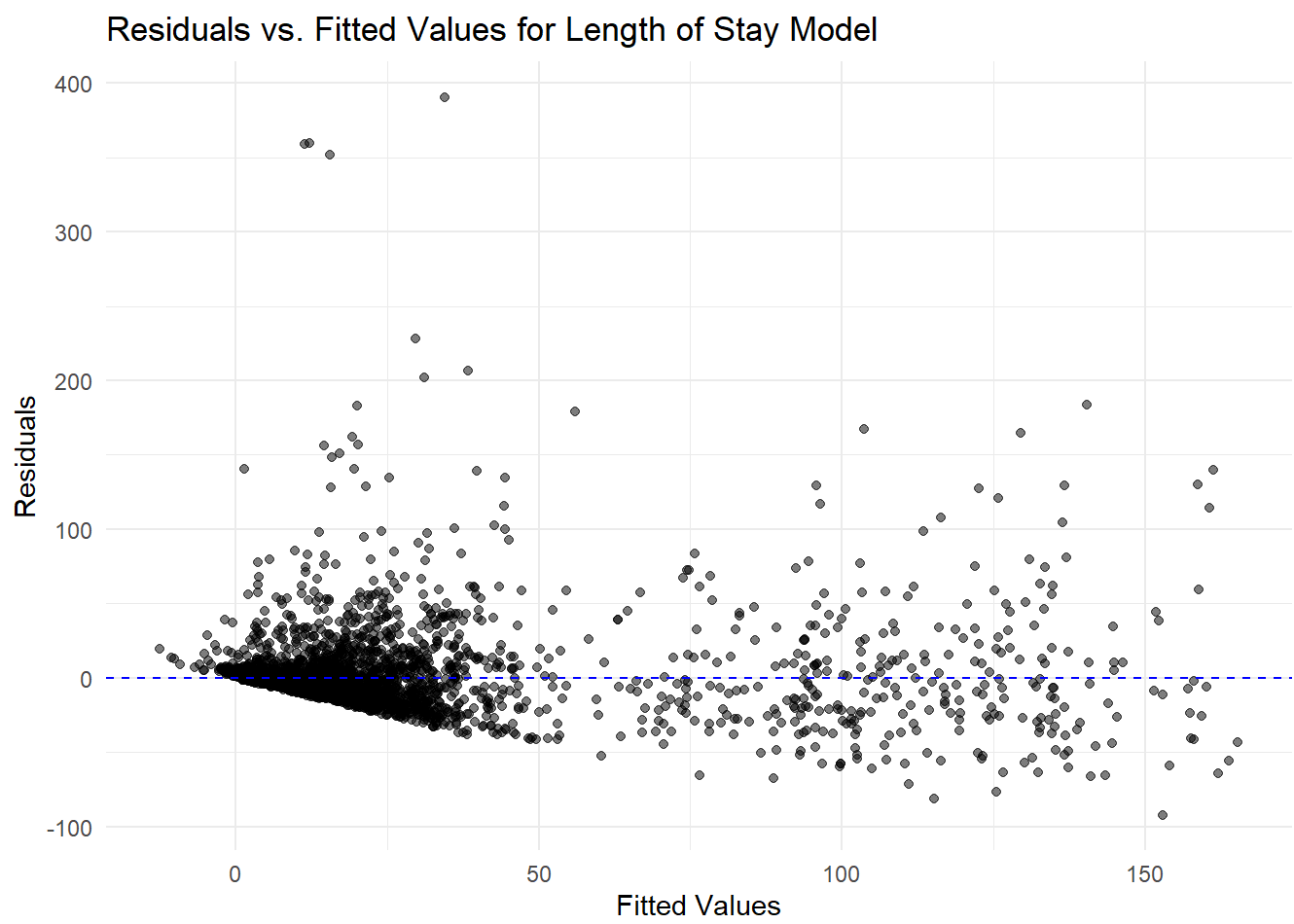
Models

```
## Linear Regression
##
## 3437 samples
## 10 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 3094, 3094, 3092, 3092, 3093, 3094, ...
## Resampling results:
##
## RMSE      Rsquared  MAE
## 27.0561  0.5489715  14.79569
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

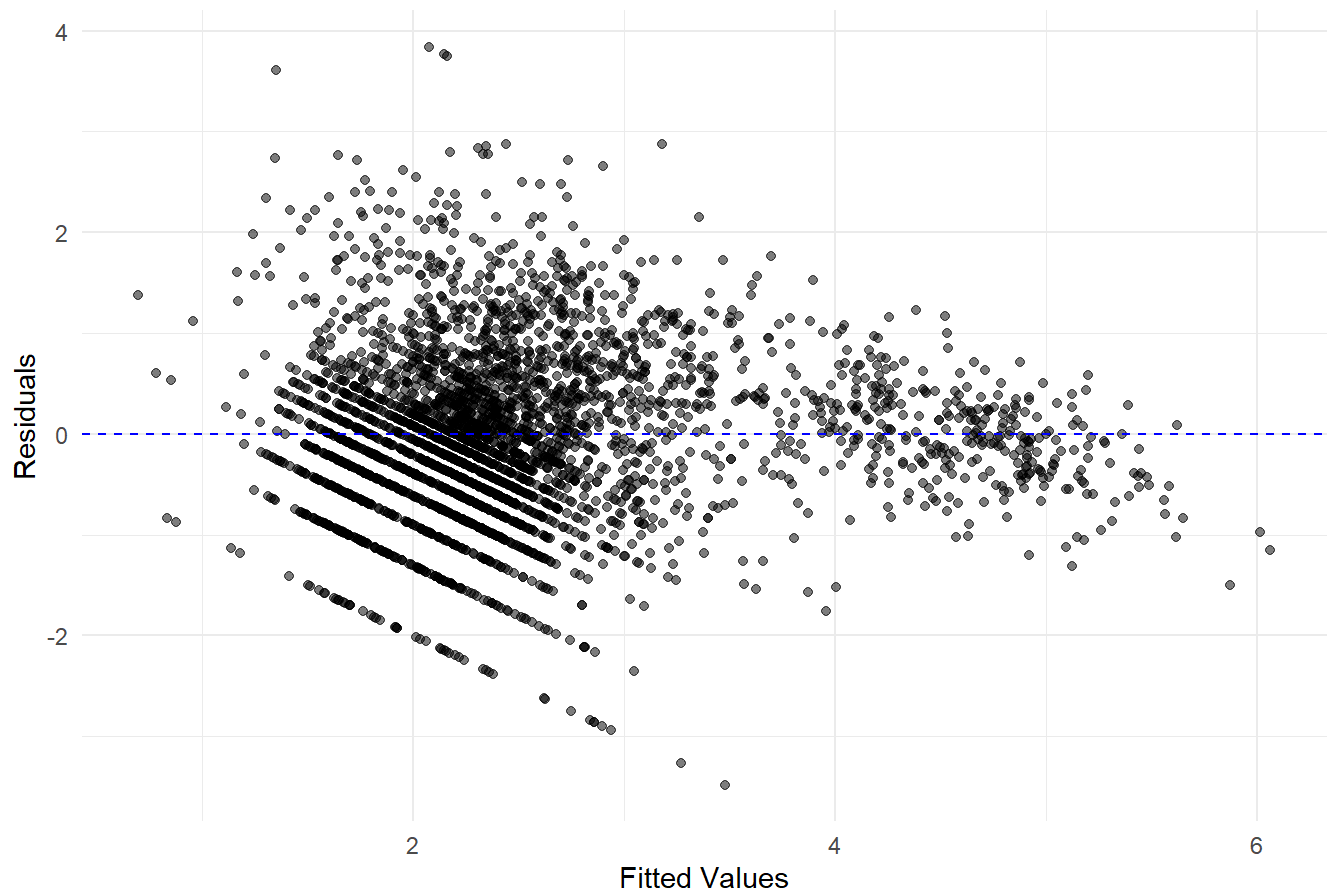
```
## Linear Regression
##
## 3437 samples
## 10 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 3094, 3093, 3092, 3094, 3093, 3094, ...
## Resampling results:
##
## RMSE      Rsquared  MAE
## 0.8291185  0.4964357  0.6325533
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```
## Linear Regression
##
## 3437 samples
## 10 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 3093, 3094, 3093, 3093, 3094, 3093, ...
## Resampling results:
##
## RMSE      Rsquared  MAE
## 0.8281494  0.4978149  0.6315676
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

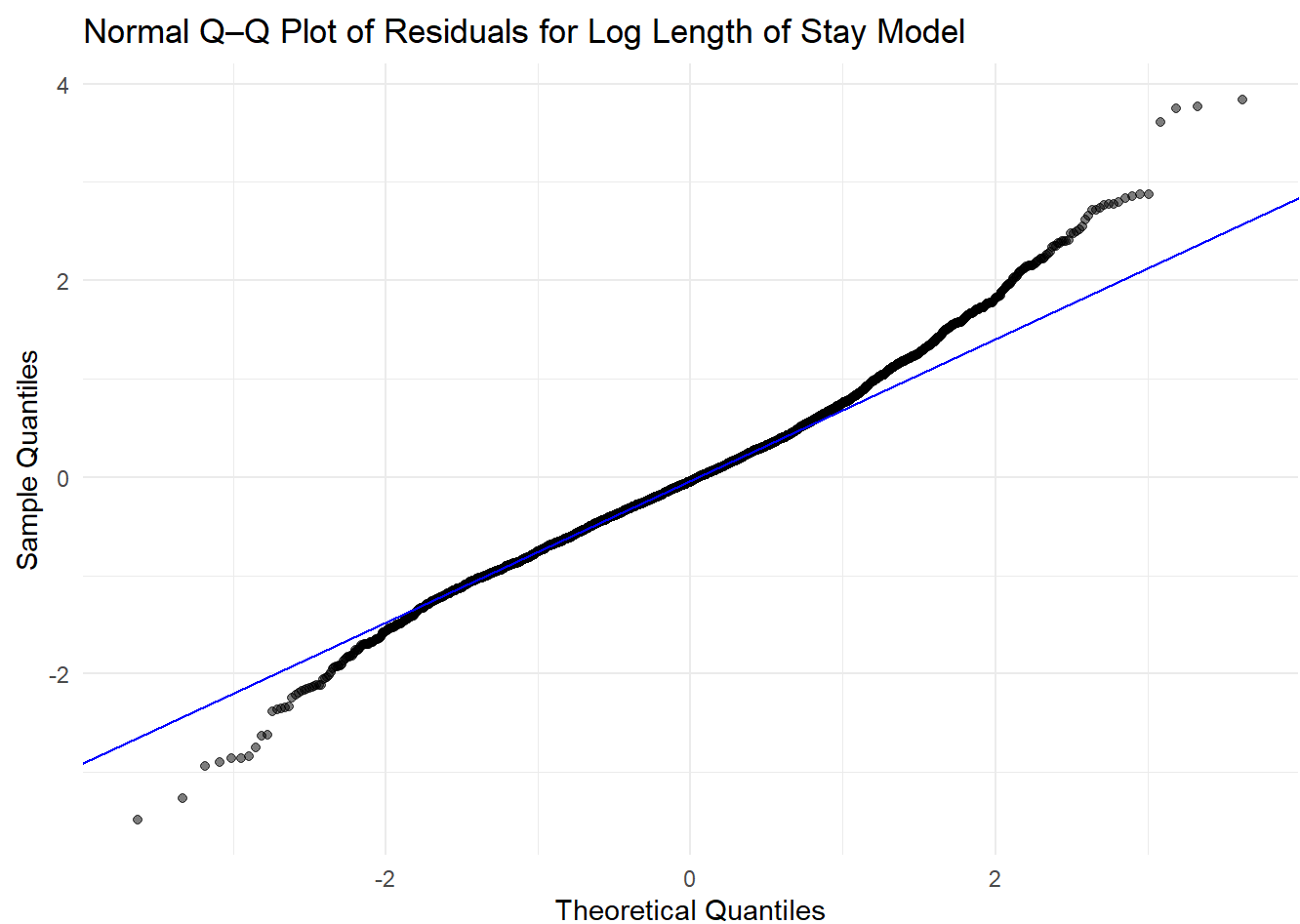
Residuals vs. Fitted Plot for Full and Log Linear Regression Models



Residuals vs. Fitted Values for Log Length of Stay Model



Normal Q-Q Plot for Log-Linear Regression Model



Decision Tree Model

```
## Call:
## rpart(formula = predictor_formula_log, data = train_data_2, method = "anova")
##   n= 3437
##
##           CP nsplit rel error   xerror   xstd
## 1 0.33344185      0 1.0000000 1.0013345 0.02203227
## 2 0.06205392      1 0.6665581 0.6784324 0.01906068
## 3 0.02685707      2 0.6045042 0.6077683 0.01726408
## 4 0.02183542      3 0.5776472 0.5895350 0.01719069
## 5 0.01893896      4 0.5558117 0.5848163 0.01725406
## 6 0.01000000      5 0.5368728 0.5429548 0.01570837
##
## Variable importance
##   birth_weight gestational_age      rop      pda      ivh
##           35           27           19           8           4
## survival_status      inborn      nec
##           2           2           2
##
## Node number 1: 3437 observations,   complexity param=0.3334419
##   mean=2.528119, MSE=1.367791
##   left son=2 (2774 obs) right son=3 (663 obs)
##   Primary splits:
##     birth_weight < 1305   to the right, improve=0.33344190, (0 missing)
##     gestational_age < 29.5 to the right, improve=0.33018800, (0 missing)
##     rop           splits as LRRRR,      improve=0.32436980, (0 missing)
##     pda           splits as LRRRLL,     improve=0.13367680, (0 missing)
##     ivh           splits as RRRRLL,     improve=0.08635687, (0 missing)
##   Surrogate splits:
##     gestational_age < 29.5 to the right, agree=0.954, adj=0.763, (0 split)
##     rop           splits as LRRRR,      agree=0.894, adj=0.449, (0 split)
##     pda           splits as LRRRLL,     agree=0.857, adj=0.261, (0 split)
##     ivh           splits as RRRRLL,     agree=0.828, adj=0.107, (0 split)
##     nec           splits as RLRL,       agree=0.815, adj=0.042, (0 split)
##
## Node number 2: 2774 observations,   complexity param=0.02685707
##   mean=2.19796, MSE=0.8606117
##   left son=4 (1736 obs) right son=5 (1038 obs)
##   Primary splits:
##     birth_weight < 2242.5 to the right, improve=0.05288640, (0 missing)
##     gestational_age < 31.5 to the right, improve=0.04643471, (0 missing)
##     inborn       splits as RL,          improve=0.03304518, (0 missing)
##     ivh          splits as RRRRLL,     improve=0.02958562, (0 missing)
##     pda          splits as L-RRLL,     improve=0.02511599, (0 missing)
##   Surrogate splits:
##     gestational_age < 34.5 to the right, agree=0.836, adj=0.561, (0 split)
##     pda           splits as L-RRLL,     agree=0.635, adj=0.025, (0 split)
```

```

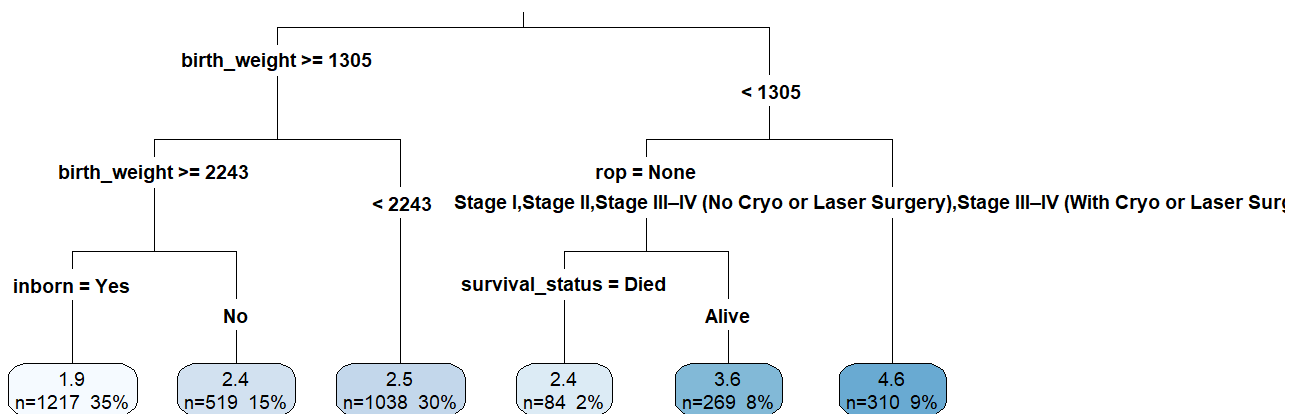
##      ivh                splits as  RRLRLL,      agree=0.634, adj=0.023, (0 split)
##      survival_status splits as  LR,            agree=0.630, adj=0.013, (0 split)
##      rop                splits as  LRRLL-,      agree=0.629, adj=0.009, (0 split)
##
## Node number 3: 663 observations,      complexity param=0.06205392
## mean=3.909509, MSE=1.125516
## left son=6 (353 obs) right son=7 (310 obs)
## Primary splits:
##      rop                splits as  LRRRR,      improve=0.39093370, (0 missing)
##      survival_status splits as  RL,            improve=0.23577600, (0 missing)
##      gestational_age < 28.5 to the right, improve=0.10290220, (0 missing)
##      birth_weight < 1055 to the right, improve=0.06557655, (0 missing)
##      pda                splits as  LRRLL-,      improve=0.03639060, (0 missing)
## Surrogate splits:
##      gestational_age < 27.5 to the right, agree=0.713, adj=0.387, (0 split)
##      birth_weight < 867 to the right, agree=0.671, adj=0.297, (0 split)
##      survival_status splits as  RL,            agree=0.581, adj=0.103, (0 split)
##      pda                splits as  LRRLL-,      agree=0.573, adj=0.087, (0 split)
##      ivh                splits as  RRRRL-,      agree=0.564, adj=0.068, (0 split)
##
## Node number 4: 1736 observations,      complexity param=0.02183542
## mean=2.032992, MSE=0.843638
## left son=8 (1217 obs) right son=9 (519 obs)
## Primary splits:
##      inborn              splits as  RL,            improve=0.07008980, (0 missing)
##      pda                splits as  L-RRRL,      improve=0.02279997, (0 missing)
##      ivh                splits as  RRLRLL,      improve=0.01969730, (0 missing)
##      gestational_age < 33.5 to the right, improve=0.01468699, (0 missing)
##      nec                splits as  RLRL,        improve=0.01423850, (0 missing)
## Surrogate splits:
##      ivh                splits as  LRRRLL,      agree=0.704, adj=0.010, (0 split)
##      gestational_age < 29.5 to the right, agree=0.702, adj=0.004, (0 split)
##      survival_status splits as  LR,            agree=0.702, adj=0.002, (0 split)
##      rop                splits as  LR-L-,      agree=0.702, adj=0.002, (0 split)
##      pda                splits as  L-RLLL,      agree=0.702, adj=0.002, (0 split)
##
## Node number 5: 1038 observations
## mean=2.47386, MSE=0.7673638
##
## Node number 6: 353 observations,      complexity param=0.01893896
## mean=3.287895, MSE=1.10688
## left son=12 (84 obs) right son=13 (269 obs)
## Primary splits:
##      survival_status splits as  RL,            improve=0.22786630, (0 missing)
##      birth_weight < 617.5 to the left, improve=0.13757610, (0 missing)
##      gestational_age < 23.5 to the left, improve=0.10787230, (0 missing)
##      pda                splits as  LLRLR-,      improve=0.02899885, (0 missing)
##      race_ethnicity splits as  RLL,            improve=0.02503565, (0 missing)
## Surrogate splits:

```

```

##      gestational_age < 24.5   to the left,  agree=0.873, adj=0.464, (0 split)
##      birth_weight    < 695    to the left,  agree=0.861, adj=0.417, (0 split)
##      ivh              splits as  RLLLR-,    agree=0.813, adj=0.214, (0 split)
##      nec              splits as  RRL-,      agree=0.799, adj=0.155, (0 split)
##
## Node number 7: 310 observations
##   mean=4.617347, MSE=0.205701
##
## Node number 8: 1217 observations
##   mean=1.874195, MSE=0.7255278
##
## Node number 9: 519 observations
##   mean=2.405356, MSE=0.922809
##
## Node number 12: 84 observations
##   mean=2.38917, MSE=1.656577
##
## Node number 13: 269 observations
##   mean=3.568538, MSE=0.6042459

```



Full Random Forest Model

```
## Ranger result
##
## Call:
## ranger(formula = predictor_formula, data = train_data, num.trees = 500,      importan
ce = "impurity")
##
## Type:                      Regression
## Number of trees:           500
## Sample size:               3437
## Number of independent variables: 10
## Mtry:                      3
## Target node size:          5
## Variable importance mode:   impurity
## Splitrule:                 variance
## OOB prediction error (MSE): 697.1425
## R squared (OOB):           0.5697649
```

```
## [1] 26.40346
```

Log-Random Forest Model

```
## Ranger result
##
## Call:
## ranger(formula = predictor_formula_log, data = train_data_2,      num.trees = 500, im
portance = "impurity")
##
## Type:                      Regression
## Number of trees:           500
## Sample size:               3437
## Number of independent variables: 10
## Mtry:                      3
## Target node size:          5
## Variable importance mode:   impurity
## Splitrule:                 variance
## OOB prediction error (MSE): 0.6393325
## R squared (OOB):           0.5327162
```

```
## [1] 0.7995827
```

Random Forest Model (Categorical Morbidities)

```
## Ranger result
##
## Call:
## ranger(formula = predictor_formula_binary, data = train_data,      num.trees = 500, i
importance = "impurity")
##
## Type:                      Regression
## Number of trees:           500
## Sample size:               3437
## Number of independent variables: 10
## Mtry:                      3
## Target node size:          5
## Variable importance mode:   impurity
## Splitrule:                 variance
## OOB prediction error (MSE): 702.9162
## R squared (OOB):           0.5662017
```

```
## [1] 26.51257
```

Log-Random Forest Model (Categorical Morbidities)

```
## Ranger result
##
## Call:
## ranger(formula = predictor_formula_log_binary, data = train_data_2,      num.trees =
500, importance = "impurity")
##
## Type:                      Regression
## Number of trees:           500
## Sample size:               3437
## Number of independent variables: 10
## Mtry:                      3
## Target node size:          5
## Variable importance mode:   impurity
## Splitrule:                 variance
## OOB prediction error (MSE): 0.6364179
## R squared (OOB):           0.5348464
```

```
## [1] 0.797758
```

Random Forest Comparison (Using OOB RMSE)

```
## # A tibble: 4 × 2
##   model                oob_rmse
##   <chr>                <dbl>
## 1 Full Random Forest    26.4
## 2 Log-Random Forest     0.800
## 3 Random Forest (Categorical) 26.5
## 4 Log-Random Forest (Categorical) 0.798
```

Hyperparameter Tuning: mtry Selection

```
## [1] 24
```

```
## # A tibble: 7 × 2
##   mtry oob_rmse
##   <dbl>   <dbl>
## 1     4    26.7
## 2     5    26.9
## 3     6    27.1
## 4     7    27.3
## 5     8    27.5
## 6     9    27.6
## 7    10    27.9
```

Grid Search for mtry and Number of Trees

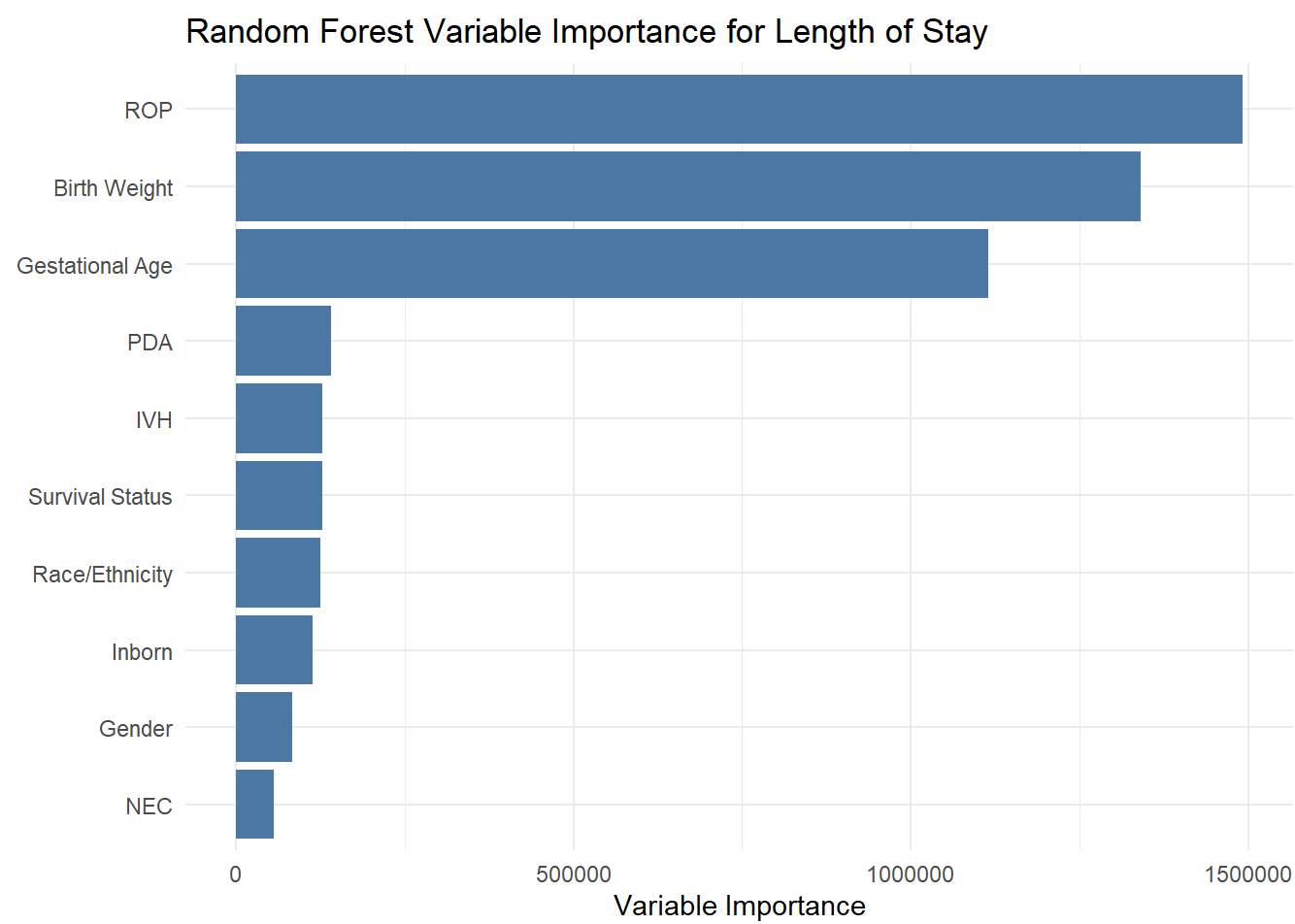
```
##   mtry num_trees
## 1    4       300
## 2    5       300
## 3    6       300
## 4    4       500
## 5    5       500
## 6    6       500
## 7    4       800
## 8    5       800
## 9    6       800
```

```
## # A tibble: 9 × 3
##   mtry num_trees oob_rmse
##   <dbl>     <dbl>   <dbl>
## 1     4       300    26.7
## 2     5       300    26.9
## 3     6       300    27.1
## 4     4       500    26.6
## 5     5       500    26.9
## 6     6       500    27.1
## 7     4       800    26.7
## 8     5       800    26.9
## 9     6       800    27.1
```

Final Tuned Random Forest Model

```
## Ranger result
##
## Call:
## ranger(formula = rf_formula, data = train_data, num.trees = 500,      mtry = 4, impor
tance = "impurity")
##
## Type:                                Regression
## Number of trees:                     500
## Sample size:                         3437
## Number of independent variables:     10
## Mtry:                                4
## Target node size:                    5
## Variable importance mode:             impurity
## Splitrule:                           variance
## OOB prediction error (MSE):           710.5583
## R squared (OOB):                     0.5614854
```

Variable Importance in the Final Random Forest Model



90th Percentile Restriction Split

##	90%
##	73

Log-Linear Regression with Restriction to the 90th Percentile Model

```
##
## Call:
## lm(formula = predictor_formula_log, data = train_data_90_log)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4312 -0.5011 -0.0226  0.4709  2.8698
##
## Coefficients: (2 not defined because of singularities)
##
##              Estimate Std. Error t value
## (Intercept)      6.172e+00  2.479e-01  24.893
## survival_statusDied -1.016e+00  7.724e-02 -13.159
## gestational_age    -5.892e-02  7.228e-03  -8.152
## birth_weight     -1.820e-04  2.826e-05  -6.442
## genderMale        1.072e-02  2.859e-02   0.375
## inbornYes        -4.025e-01  3.463e-02 -11.625
## race_ethnicityOther  6.374e-02  4.107e-02   1.552
## race_ethnicityWhite  1.117e-01  3.101e-02   3.604
## ivhGrade II      -2.551e-02  2.078e-01  -0.123
## ivhGrade III     -2.385e-01  2.500e-01  -0.954
## ivhGrade IV      -2.872e-01  1.767e-01  -1.625
## ivhNone          -4.763e-01  7.755e-02  -6.141
## ivhMissing       -1.236e+00  3.542e-01  -3.490
## ropStage I        8.036e-01  1.249e-01   6.436
## ropStage II       8.480e-01  1.364e-01   6.215
## ropStage III-IV (No Cryo or Laser Surgery)  9.523e-01  5.577e-01   1.708
## ropStage III-IV (With Cryo or Laser Surgery) 8.388e-01  3.522e-01   2.381
## necNone          -7.462e-01  1.347e-01  -5.538
## necSurgical      -1.484e-01  2.230e-01  -0.665
## necMissing              NA              NA              NA
## pdaNone (Medication Only)  2.703e-01  3.951e-01   0.684
## pdaYes (Ligated)      4.638e-01  1.355e-01   3.423
## pdaYes (Medically Treated)  4.196e-01  8.406e-02   4.992
## pdaYes (Untreated)    3.689e-01  4.186e-02   8.814
## pdaMissing              NA              NA              NA
##
##              Pr(>|t|)
## (Intercept)      < 2e-16 ***
## survival_statusDied < 2e-16 ***
## gestational_age    5.17e-16 ***
## birth_weight     1.37e-10 ***
## genderMale        0.707654
## inbornYes        < 2e-16 ***
## race_ethnicityOther 0.120703
## race_ethnicityWhite 0.000318 ***
## ivhGrade II       0.902295
## ivhGrade III      0.340005
```

```
## ivhGrade IV 0.104205
## ivhNone 9.25e-10 ***
## ivhMissing 0.000490 ***
## ropStage I 1.42e-10 ***
## ropStage II 5.81e-10 ***
## ropStage III-IV (No Cryo or Laser Surgery) 0.087790 .
## ropStage III-IV (With Cryo or Laser Surgery) 0.017303 *
## necNone 3.33e-08 ***
## necSurgical 0.505950
## necMissing NA
## pdaNone (Medication Only) 0.494039
## pdaYes (Ligated) 0.000627 ***
## pdaYes (Medically Treated) 6.32e-07 ***
## pdaYes (Untreated) < 2e-16 ***
## pdaMissing NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7805 on 3078 degrees of freedom
## Multiple R-squared:  0.3231, Adjusted R-squared:  0.3182
## F-statistic: 66.77 on 22 and 3078 DF,  p-value: < 2.2e-16
```

Final Random Forest with Restriction to the 90th Percentile Model

```
## Ranger result
##
## Call:
## ranger(formula = rf_formula, data = train_data_90, num.trees = 500, mtry = 4, im
portance = "impurity")
##
## Type: Regression
## Number of trees: 500
## Sample size: 3101
## Number of independent variables: 10
## Mtry: 4
## Target node size: 5
## Variable importance mode: impurity
## Splitrule: variance
## OOB prediction error (MSE): 155.7376
## R squared (OOB): 0.3911426
```

Random Forest Classification Model (5 Categories)

```
##
## 0-5 6-10 11-20 21-30 31+
## 1339 991 696 283 988
```

```
## Rows: 4,297
## Columns: 17
## $ patient_id      <dbl> 1384, 4010, 2751, 1035, 1117, 2180, 2608, 603, 435, 16...
## $ survival_status <fct> Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive...
## $ gestational_age <dbl> 34, 30, 30, 33, 27, 26, 38, 32, 32, 30, 30, 38, 31, 35...
## $ birth_weight    <dbl> 2700, 1250, 1380, 2190, 910, 830, 2850, 1900, 1520, 12...
## $ gender          <fct> Male, Female, Male, Female, Male, Male, Female, Female...
## $ inborn          <fct> Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, ...
## $ race_ethnicity  <fct> Other, White, White, White, White, Black, White, Black...
## $ length_of_stay  <dbl> 15, 5, 5, 3, 52, 46, 15, 5, 5, 52, 52, 15, 33, 10, 3, ...
## $ ivh             <fct> None, None, None, None, None, None, None, None, None, ...
## $ rop             <fct> None, None, None, None, Stage I, None, None, None, Non...
## $ nec             <fct> None, None, None, None, Medical, None, None, None, Non...
## $ pda             <fct> None, None (Medication Only), None, None, Yes (Ligated...
## $ ivh_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ rop_2           <fct> No, No, No, No, Yes, No, No, No, No, Yes, No, No, No, ...
## $ nec_2           <fct> No, No, No, No, Yes, No, No, No, No, No, No, No, No, N...
## $ pda_2           <fct> No, No, No, No, Yes, Yes, No, No, No, Yes, Yes, No, Ye...
## $ los_category    <fct> 11-20, 0-5, 0-5, 0-5, 31+, 31+, 11-20, 0-5, 0-5, 31+, ...
```

```
## Rows: 3,436
## Columns: 17
## $ patient_id      <dbl> 603, 435, 1241, 3783, 2123, 3584, 904, 3773, 2031, 349...
## $ survival_status <fct> Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive...
## $ gestational_age <dbl> 32, 32, 31, 35, 33, 36, 40, 31, 36, 39, 40, 39, 35, 34...
## $ birth_weight    <dbl> 1900, 1520, 1630, 2710, 1930, 4110, 4570, 1700, 2620, ...
## $ gender          <fct> Female, Female, Female, Female, Male, Male, Female, Ma...
## $ inborn          <fct> Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, ...
## $ race_ethnicity  <fct> Black, Black, White, Other, White, Black, Black, Black...
## $ length_of_stay  <dbl> 5, 5, 3, 2, 5, 3, 1, 5, 5, 3, 2, 3, 3, 5, 3, 3, 4, 26,...
## $ ivh             <fct> None, None, None, None, None, None, None, None, None, ...
## $ rop             <fct> None, None, None, None, None, None, None, None, None, ...
## $ nec             <fct> None, None, None, None, None, None, None, None, None, ...
## $ pda             <fct> None, None, None, None, None, None, None, None, None, ...
## $ ivh_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ rop_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ nec_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ pda_2           <fct> No, No, No, No, No, No, No, No, No, Yes, No, No, No, N...
## $ los_category    <fct> 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, ...
```

```
## Rows: 861
## Columns: 17
## $ patient_id      <dbl> 4010, 2751, 1035, 2980, 396, 428, 896, 2890, 3372, 381...
## $ survival_status <fct> Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive...
## $ gestational_age <dbl> 30, 30, 33, 30, 31, 33, 34, 37, 35, 38, 37, 35, 23, 26...
## $ birth_weight    <dbl> 1250, 1380, 2190, 680, 1650, 1460, 2795, 3205, 2095, 2...
## $ gender          <fct> Female, Male, Female, Female, Male, Female, Male, Male...
## $ inborn          <fct> Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, ...
## $ race_ethnicity  <fct> White, White, White, White, White, Black, White, Black...
## $ length_of_stay  <dbl> 5, 5, 3, 52, 3, 9, 7, 3, 15, 5, 5, 5, 7, 1, 4, 12, 23,...
## $ ivh             <fct> None, None, None, None, None, None, None, None, None, ...
## $ rop             <fct> None, None, None, None, None, None, None, None, None, ...
## $ nec             <fct> None, None, None, None, None, None, None, None, None, ...
## $ pda             <fct> None (Medication Only), None, None, Yes (Untreated), N...
## $ ivh_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, Yes, No, N...
## $ rop_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ nec_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, Yes, N...
## $ pda_2           <fct> No, No, No, Yes, No, No, No, Yes, No, No, No, Yes, Yes...
## $ los_category    <fct> 0-5, 0-5, 0-5, 31+, 0-5, 6-10, 6-10, 0-5, 11-20, 0-5, ...
```

```
##      mtry_2 num_trees_2
## 1         3         300
## 2         4         300
## 3         5         300
## 4         6         300
## 5         7         300
## 6         8         300
## 7         3         500
## 8         4         500
## 9         5         500
## 10        6         500
## 11        7         500
## 12        8         500
## 13        3         800
## 14        4         800
## 15        5         800
## 16        6         800
## 17        7         800
## 18        8         800
```

```
## # A tibble: 18 × 4
##   mtry_2 num_trees_2 oob_error oob_accuracy
##   <dbl>      <dbl>      <dbl>      <dbl>
## 1      3          300      0.454      0.546
## 2      4          300      0.453      0.547
## 3      5          300      0.453      0.547
## 4      6          300      0.456      0.544
## 5      7          300      0.458      0.542
## 6      8          300      0.461      0.539
## 7      3          500      0.455      0.545
## 8      4          500      0.452      0.548
## 9      5          500      0.453      0.547
## 10     6          500      0.455      0.545
## 11     7          500      0.458      0.542
## 12     8          500      0.461      0.539
## 13     3          800      0.455      0.545
## 14     4          800      0.452      0.548
## 15     5          800      0.453      0.547
## 16     6          800      0.455      0.545
## 17     7          800      0.458      0.542
## 18     8          800      0.460      0.540
```

```
## # A tibble: 1 × 4
##   mtry_2 num_trees_2 oob_error oob_accuracy
##   <dbl>      <dbl>      <dbl>      <dbl>
## 1      4          500      0.452      0.548
```

```

## Confusion Matrix and Statistics
##
##           Reference
## Prediction 0-5 6-10 11-20 21-30 31+
##      0-5   194   110    68    12   20
##      6-10    55    58    32     7   19
##     11-20    14    18    16     9   15
##     21-30     0     1     2     2    4
##     31+    10    16    22    13  144
##
## Overall Statistics
##
##           Accuracy : 0.4808
##           95% CI : (0.447, 0.5148)
##    No Information Rate : 0.3171
##    P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.2931
##
## Mcnemar's Test P-Value : 1.087e-14
##
## Statistics by Class:
##
##           Class: 0-5 Class: 6-10 Class: 11-20 Class: 21-30
## Sensitivity          0.7106      0.28571      0.11429      0.046512
## Specificity          0.6429      0.82827      0.92233      0.991443
## Pos Pred Value       0.4802      0.33918      0.22222      0.222222
## Neg Pred Value       0.8271      0.78986      0.84284      0.951878
## Prevalence           0.3171      0.23577      0.16260      0.049942
## Detection Rate       0.2253      0.06736      0.01858      0.002323
## Detection Prevalence 0.4692      0.19861      0.08362      0.010453
## Balanced Accuracy     0.6767      0.55699      0.51831      0.518977
##
##           Class: 31+
## Sensitivity          0.7129
## Specificity          0.9074
## Pos Pred Value       0.7024
## Neg Pred Value       0.9116
## Prevalence           0.2346
## Detection Rate       0.1672
## Detection Prevalence 0.2381
## Balanced Accuracy     0.8102

```

Random Forest Classification Model (3 Categories)

```
##
## 0-5 6-20 21+
## 1339 1687 1271
```

```
## Rows: 4,297
## Columns: 17
## $ patient_id      <dbl> 1384, 4010, 2751, 1035, 1117, 2180, 2608, 603, 435, 16...
## $ survival_status <fct> Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive...
## $ gestational_age <dbl> 34, 30, 30, 33, 27, 26, 38, 32, 32, 30, 30, 38, 31, 35...
## $ birth_weight    <dbl> 2700, 1250, 1380, 2190, 910, 830, 2850, 1900, 1520, 12...
## $ gender          <fct> Male, Female, Male, Female, Male, Male, Female, Female...
## $ inborn          <fct> Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes,...
## $ race_ethnicity  <fct> Other, White, White, White, White, Black, White, Black...
## $ length_of_stay  <dbl> 15, 5, 5, 3, 52, 46, 15, 5, 5, 52, 52, 15, 33, 10, 3, ...
## $ ivh             <fct> None, None, None, None, None, None, None, None, None, ...
## $ rop             <fct> None, None, None, None, Stage I, None, None, None, Non...
## $ nec             <fct> None, None, None, None, Medical, None, None, None, Non...
## $ pda             <fct> None, None (Medication Only), None, None, Yes (Ligated...
## $ ivh_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ rop_2           <fct> No, No, No, No, Yes, No, No, No, No, Yes, No, No, No, ...
## $ nec_2           <fct> No, No, No, No, Yes, No, No, No, No, No, No, No, No, N...
## $ pda_2           <fct> No, No, No, No, Yes, Yes, No, No, No, Yes, Yes, No, Ye...
## $ los_category_2  <fct> 6-20, 0-5, 0-5, 0-5, 21+, 21+, 6-20, 0-5, 0-5, 21+, 21...
```

```
## Rows: 3,436
## Columns: 17
## $ patient_id      <dbl> 4010, 2751, 1035, 603, 1241, 396, 3783, 2123, 2890, 35...
## $ survival_status <fct> Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive...
## $ gestational_age <dbl> 30, 30, 33, 32, 31, 31, 35, 33, 37, 36, 40, 37, 26, 36...
## $ birth_weight    <dbl> 1250, 1380, 2190, 1900, 1630, 1650, 2710, 1930, 3205, ...
## $ gender          <fct> Female, Male, Female, Female, Female, Male, Female, Ma...
## $ inborn          <fct> Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes, Yes,...
## $ race_ethnicity  <fct> White, White, White, Black, White, White, Other, White...
## $ length_of_stay  <dbl> 5, 5, 3, 5, 3, 3, 2, 5, 3, 3, 1, 5, 1, 5, 3, 3, 3, 4, ...
## $ ivh             <fct> None, None, None, None, None, None, None, None, None, ...
## $ rop             <fct> None, None, None, None, None, None, None, None, None, ...
## $ nec             <fct> None, None, None, None, None, None, None, None, None, ...
## $ pda             <fct> None (Medication Only), None, None, None, None, None, ...
## $ ivh_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ rop_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ nec_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ pda_2           <fct> No, No, No, No, No, No, No, No, Yes, No, No, No, No, N...
## $ los_category_2  <fct> 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, 0-5, ...
```

```
## Rows: 861
## Columns: 17
## $ patient_id      <dbl> 2608, 435, 428, 3816, 3149, 3773, 1789, 199, 1718, 274...
## $ survival_status <fct> Alive, Alive, Alive, Alive, Alive, Alive, Alive, Alive...
## $ gestational_age <dbl> 38, 32, 33, 38, 35, 31, 35, 38, 35, 40, 39, 34, 34, 33...
## $ birth_weight    <dbl> 2850, 1520, 1460, 2680, 1815, 1700, 2520, 3195, 2465, ...
## $ gender          <fct> Female, Female, Female, Male, Female, Male, Female, Fe...
## $ inborn          <fct> Yes, Yes, Yes, Yes, Yes, Yes, No, Yes, No, Yes, Yes, Y...
## $ race_ethnicity  <fct> White, Black, Black, Other, White, Black, White, Black...
## $ length_of_stay  <dbl> 15, 5, 9, 5, 5, 5, 4, 12, 11, 2, 3, 5, 8, 3, 1, 6, 26,...
## $ ivh             <fct> None, None, None, None, Grade I, None, None, None, Non...
## $ rop             <fct> None, None, None, None, None, None, None, None, None, ...
## $ nec             <fct> None, None, None, None, None, None, None, None, None, ...
## $ pda             <fct> None, None, None, None, Yes (Untreated), None, None, N...
## $ ivh_2           <fct> No, No, No, No, Yes, No, No, No, No, No, No, No, No, N...
## $ rop_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ nec_2           <fct> No, No, No, No, No, No, No, No, No, No, No, No, No, No...
## $ pda_2           <fct> No, No, No, No, Yes, No, No, No, No, No, No, No, No, N...
## $ los_category_2  <fct> 6-20, 0-5, 6-20, 0-5, 0-5, 0-5, 0-5, 6-20, 6-20, 0-5, ...
```

```
##      mtry_3 num_trees_3
## 1         3         300
## 2         4         300
## 3         5         300
## 4         6         300
## 5         7         300
## 6         8         300
## 7         3         500
## 8         4         500
## 9         5         500
## 10        6         500
## 11        7         500
## 12        8         500
## 13        3         800
## 14        4         800
## 15        5         800
## 16        6         800
## 17        7         800
## 18        8         800
```

```
## # A tibble: 18 × 4
##   mtry_3 num_trees_3 oob_error oob_accuracy
##   <dbl>      <dbl>      <dbl>      <dbl>
## 1      3          300      0.301      0.699
## 2      4          300      0.299      0.701
## 3      5          300      0.301      0.699
## 4      6          300      0.304      0.696
## 5      7          300      0.305      0.695
## 6      8          300      0.309      0.691
## 7      3          500      0.301      0.699
## 8      4          500      0.300      0.700
## 9      5          500      0.300      0.700
## 10     6          500      0.303      0.697
## 11     7          500      0.307      0.693
## 12     8          500      0.309      0.691
## 13     3          800      0.301      0.699
## 14     4          800      0.298      0.702
## 15     5          800      0.300      0.700
## 16     6          800      0.302      0.698
## 17     7          800      0.306      0.694
## 18     8          800      0.309      0.691
```

```
## # A tibble: 1 × 4
##   mtry_3 num_trees_3 oob_error oob_accuracy
##   <dbl>      <dbl>      <dbl>      <dbl>
## 1      4          800      0.298      0.702
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction 0-5 6-20 21+
##           0-5 121   85  17
##           6-20 141  219  83
##           21+   6   34 155
##
## Overall Statistics
##
##           Accuracy : 0.5749
##           95% CI : (0.5411, 0.6082)
##           No Information Rate : 0.3926
##           P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.3463
##
##           Mcnemar's Test P-Value : 1.259e-08
##
## Statistics by Class:
##
##           Class: 0-5 Class: 6-20 Class: 21+
## Sensitivity           0.4515      0.6479      0.6078
## Specificity           0.8280      0.5717      0.9340
## Pos Pred Value        0.5426      0.4944      0.7949
## Neg Pred Value        0.7696      0.7153      0.8498
## Prevalence            0.3113      0.3926      0.2962
## Detection Rate        0.1405      0.2544      0.1800
## Detection Prevalence  0.2590      0.5145      0.2265
## Balanced Accuracy      0.6397      0.6098      0.7709
```

Predictive Performance

```
## # A tibble: 4 × 2
##   Model                                RMSE
##   <chr>                                <dbl>
## 1 Log-Linear (Full Test)                31.5
## 2 Log-Linear (90th Percentile)         12.6
## 3 Random Forest (Full Test)            29.5
## 4 Random Forest (90th Percentile)      11.3
```

```
## [1] 33.15052
```